AutoBench Service — Developer Guide

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> Hand-maintained, unlike the generated `results/12run-*.md` files which stamp themselves. Bump the line above when you edit this guide.

A task-oriented guide to driving the AutoBench Service over its RESTful API. Every `curl` below is derived from the flows we exercised end-to-end (gsm8k single-turn, tau2 multi-turn) on OpenShift (`ykt3`) and KinD (`kind-rossoctl`).

- **One Service, many clusters.** The issuer of your token selects the instance, and with it the cluster your benchmark workloads run on — every agent and MCP server in one namespace, `team1` in every example below. The layout, and the rules for sharing one Service, are in the Admin Guide, §1. The examples run where the install scripts put things, the workloads on the Service's own cluster (Admin Guide, §5).
- Reference the machine-readable contract in `openapi.json` / `openapi.yaml`.
- Installing and operating the Service — prerequisites, environment variables, the Helm chart for KinD and OpenShift — is `ADMIN_GUIDE.md`.
- Design rationale (what the Service can and cannot enact) lives in `SERVICE_DESIGN_DECISIONS.md` and `KUBECTL_DEPENDENCY_INVENTORY.md`.
- **Every host below is a placeholder on `example.com`** — this repo is public, so it names platforms rather than endpoints. Route *shapes* are real (`<service>-<namespace>.apps.<cluster>`), so substitute your own domain and the recipes work unchanged. The two exceptions are the S3 object URLs in §6.6 and §7.0, which are live and anonymously fetchable from the public bucket.

In a hurry? §3 gets you a token, §7.0 is one benchmark run start to finish as copy-pasteable `curl`, and §7.1 is the same thing as a single command.

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1. Mental model (read this first)

The Service is **pure-Python and HTTP-only**. It never calls a cluster API (no kubectl, no kubeconfig). It talks to the Rossoctl backend over HTTP(S) and to Keycloak for tokens. Two consequences shape everything below:

1. **Two independent tokens.**
    - The **caller JWT** you present in `Authorization: Bearer …` is used *only* for attribution (`preferred_username`) and routing (`iss` selects which instance config applies). It is **never**

forwarded upstream.
    - For each request the Service mints its **own** ROPC token (the instance's `benchmarker` credential, password grant) and presents *that* to Rossoctl.
2. **The Service enacts only what HTTP allows.** It deploys workloads (agents/tools), runs benchmarks, reads reports, and exports to S3. It **cannot** create cluster Secrets or overlay per-agent ConfigMaps — those are provisioned out-of-band, and the Service reports on them (see §4 onboarding and the `424` / `422` responses).

Where things run

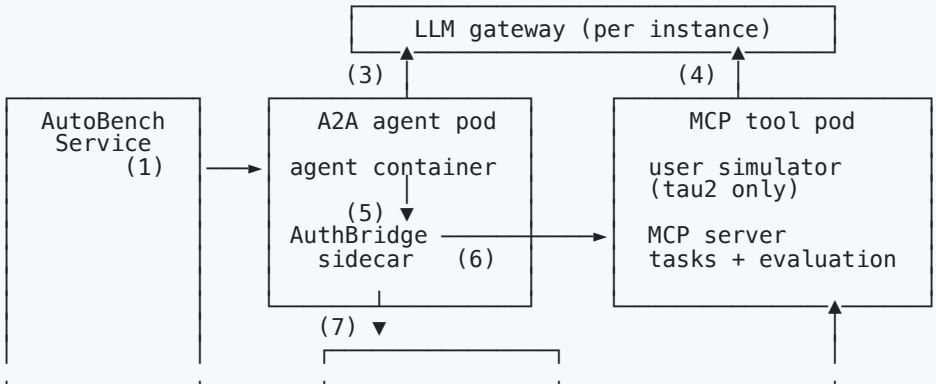
The Service sits outside the clusters it benchmarks, conceptually even when it shares one with them. Each instance (one per Keycloak issuer, §4.1) names a Keycloak, a Rossoctl backend and how to reach the workloads that Rossoctl deploys. MLflow lives with the Service, which writes its own `Agent.Session` spans there and reads every trace back. The agents' own spans travel through an OTEL collector **on the workload cluster**, which forwards them to that MLflow. S3 is external and shared by all instances. Slide 7 of `AutoBench.pptx` draws it.

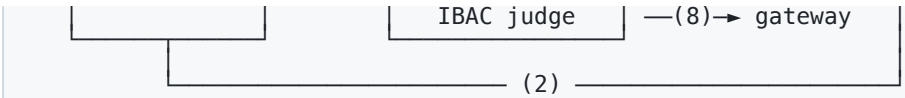
Endpoint map

Group	Endpoints	Auth
Identity	`GET /hello`, `GET /healthz` (public, unschematized)	caller JWT
Config	`GET /config`, `PUT /config`, `GET /mlflow/health`	**benchmarker only**
Discovery	`GET /namespaces`, `GET /benchmarks`, `GET /benchmarks/{name}`	caller JWT
Raw workloads	`GET/POST /agents`, `GET/DELETE /agents/{ns}/{name}`, same for `/tools`	caller JWT
Benchmark deploy	`POST /benchmarks/{name}/deploy`, `DELETE /benchmarks/{name}/deploy`, `GET /benchmarks/{name}/status`	caller JWT
Run lifecycle	`POST /benchmarks/{name}/runs`, `GET …/runs`, `GET …/runs/{run_id}`	caller JWT
Reporting	`GET …/runs/{run_id}/report`, `GET /benchmarks/{name}/report`	caller JWT

The run-time data path

The endpoint map above is the **control** plane — what you call. During a run a different set of **flows** carries the actual work, and none of them are HTTP calls you make. Slide 8 of `AutoBench.pptx` draws this; the same eight flows in text, numbered identically:





#	flow	is the sidecar on it?
1	Service → agent container: `send_prompt`, **once per task**	**yes, inbound** — logged as `a2a-parser`; `isAction=false`, so no judge call
2	Service → MCP server: `list_tasks`, `create_session`, `evaluate_session`, `delete_session`	no — a different pod, and the Service does not dial through the proxy
3	agent → LLM gateway: N chat calls per task	**no** — see below
4	MCP tool → LLM gateway: tau2's user simulator (a *second* model)	no — a different pod
5	agent → sidecar: the agent's own MCP tool calls	**this is the interception point**
6	sidecar → MCP server: once the call is authorized	outbound `mcp-parser`, `method=tools/call` `isAction=true`
7	sidecar → IBAC judge: one call per `isAction` tool call	—
8	judge → LLM gateway: the verdict is itself an LLM call	—

"Flow", never "leg", for these eight. Elsewhere in this repo — §7.5 below, `BM_ORDER`, the comparison reports, `PLUGIN_OVERHEAD.md` — a **leg** is one of the 12 parameterized *runs*. Both numberings reach 8, so "leg 6" is ambiguous where it matters: as a run it is `ibac-only`, as a flow it is sidecar → MCP server.

Three things about this shape are not guessable from the API surface:

The agent has its own MCP connection, separate from the Service's. `MCP_URL` is injected into the agent's env at deploy time (always `svc.cluster.local`, since agent→tool is intra-cluster). The Service opens and grades the session (flow 2); the agent's tool calls (flows 5–6) mutate that same session's state, correlated by `session_id` — which travels as A2A request metadata, *not* in the prompt text. So grading reads tool-side state, and the runner discards whatever the agent replies: an agent that answers in prose without making the submitting tool call fails the task.

Interception is a blanket forward proxy plus an allowlist, not a tool-aware hook. With AuthBridge enabled the operator injects `HTTP_PROXY=HTTPS_PROXY=http://127.0.0.1:8081` — the sidecar — into the agent pod, which would capture *every* outbound HTTP call it makes, inference included. What keeps flow 3 out of it is the instance config's `no_proxy`, which names the LLM gateway host (and the OTEL collector, and Keycloak). IBAC additionally sets `judge_inference: false`, so even proxied inference would not be judged. Both would have to change for the agent's model calls to be authorized.

A ready sidecar proves nothing about enforcement. Because the switch is a proxy-env allowlist, a plausible-looking config can leave the outbound plugins seeing no traffic at all: setting `workload_llm.disable_proxy: true` makes the Service inject `HTTP_PROXY=""` first, and the operator only adds a var when it is *absent*, so it skips `HTTP_PROXY` while still setting `HTTPS_PROXY`. `MCP_URL` is `http://`, so tool calls went direct — and a `no_proxy` containing `.svc.cluster.local` excluded them a second time. The sidecar was ready, the pipeline rendered, the judge was called **zero** times. Keep `disable_proxy` off and keep the *wildcards* out of `no_proxy` while keeping the named hosts in it, then verify by **counting judge completions across a run** — IBAC emits no log line of its own, so sidecar logs cannot tell you. `initialize`

and `tools/list` are `isAction=false` and legitimately never reach the judge, so a zero count only means something if real tool calls occurred. Cost figures for each layer: `PLUGIN_OVERHEAD.md`.

One task, end to end

The flows above are spatial. This is the same system in time — one task, from the Service's first call to its last, with the span each step appears as:

```
Service:  create_session(task_id)                    → MCP pod     [MCP.CreateSession]
Service:  send_prompt(task text; session_id in A2A   → agent       [Agent.Call]
          request metadata)                                        ── ONE message, once per task
  agent:    connect_mcp — tools/list                 → MCP pod     [connect_mcp]     ~40–50 ms
  agent:    create_agent                                           [create_agent]    ~12–15 ms
warm,
                                                                    up to 5.2 s on a cold pod
  agent:    initial_observation — local, no I/O                    [execute_tool
initial_observation]
                                                                    ~70 µs, never counted
  agent:    LOOP  chat(model) → execute_tool(…)      → MCP pod     [chat <model>]
                  → observation → chat → …                        [execute_tool <tool>]
  agent:    returns its final message, when the model asks for no further tool
Service:  evaluate_session                           → MCP pod     [Evaluator.Evaluate]
Service:  delete_session                             → MCP pod     (in a `finally`; no span)
```

Five things that trace makes concrete:

- **One A2A turn per task.** Everything the agent does — every model call, every tool call — happens inside that single streaming request. The agent-side `POST /` span brackets it, so `Agent.Call − POST /` is the Service's own per-task cost: a median **16.0 ms of a 5.7 s gsm8k task, 19.4 ms of a 104 s tau2 one** (v1.36 pair).
- **The Service issues no tool call.** Its MCP traffic is `list_tasks`, `create_session`, `evaluate_session`, `delete_session`. `execute_tool` is the agent's outbound call, named by the agent; the Service never learns a tool's name.
- **The tool list is fetched per task, not per pod.** `connect_mcp`, `create_agent` and `invoke_agent` each appear exactly once per task in every leg.
- **The session id is the only thing joining the two halves.** It travels in A2A request metadata, never in the prompt text, and it is why the agent's tool calls mutate the session the Service will grade.
- **The reply text is discarded.** Grading reads the session's final state, so on gsm8k the answer has to arrive through the `submit` tool call; prose alone fails the task.

Who decides what happens inside a task

The eight flows say who *talks to* whom. They do not say who **decides** — and the answer surprises people: the sequence of model calls and tool calls inside a task is **defined nowhere**. It is produced, one step at a time, by the model.

component	defines	does not define
MCP pod	the task list; each task's initial state and instruction; the **tool surface** — names, schemas, semantics, and the observations tools return; the **evaluator** (`evaluate_session` → verdict)	any ordering. It answers calls; it never asks for one
agent runtime (inside the agent image)	the **loop** — model → tool → observation → model — and when to stop	which tool, with which arguments

component	defines	does not define
the Service	one prompt per task (`runner/prompt.py`: the task text plus optional context, and no tool instructions), the session lifecycle, the per-task timeout, the telemetry	anything about the trajectory
the model	**every step**	—

Deterministic task *selection*, nondeterministic task *execution*. The runner always slices the first `max_tasks` ids off a fixed list, which is what makes legs comparable. Within a task nothing is fixed: the same task id, same model, same platform, run in two different legs, takes a different number of steps — tau2 task `9` went 12 chat / 12 tool in one leg and 11 / 11 in another; appworld `3d9a636_3` went 34 / 17 and 22 / 11. Of the tasks that appear in more than one leg, **12 of 20 tau2 tasks and 8 of 8 appworld tasks differ**, while all 20 gsm8k tasks agree — there the shape is 1 chat / 1 tool and there is nothing to vary.

How the model knows when to stop. Not from our prompt — the terminal tool's name appears nowhere in the Service. The agent fetches the tool declarations from the MCP pod at `connect_mcp`, and the benchmark's own schema descriptions are what mark one tool as the answer/finish channel. On gsm8k the entire first model call is **320 input tokens** — the runtime's system prompt, our task text and every tool schema combined — so that instruction is a terse tool description, not a protocol. In practice it lands: gsm8k ends on `submit` in **172 of 172** tasks and appworld on `finish` in **41 of 47** (the other six stopped mid-task on the agent's own defect). tau2 has **no** terminal tool at all — all **60 of 60** tasks end on `message`, a reply to the simulated customer, the loop exiting simply because the model asked for no further tool.

Two consequences worth carrying into every number downstream. Grading reads the session's **final state**, never a reference trajectory, so a task can pass by two routes at different token cost — per-task cost and latency are distributions, which is why the reports carry CV columns rather than point figures. And an appworld gap between platforms is not automatically a platform difference: the turn counts differ run to run on their own.

Counts computed over every mirrored `span_report.ndjson` from the v1.36 matrix, both platforms: per task, `counted` chat and tool spans, grouped by (benchmark, task id, model).

2. Cost and performance (which benchmark, and what a run costs)

Everything above is mechanism: what the pieces are and who decides what. This section is the money and the minutes — which benchmark answers your question, what a leg of it costs in tokens and dollars, and which of those numbers are safe to divide. Every figure here is measured from the v1.36 matrices, not estimated; the generators that compute them are in 11. Regenerating the documents.

Picking a benchmark, and what a run costs

If you want to…	use	on model	costs about
check a cluster/deploy/auth/telemetry path works	**gsm8k**, 1–10 tasks	gpt-5-mini	< $0.01
exercise concurrency and volume cheaply	**gsm8k**, 50 tasks at `max_parallel_sessions=4`	gpt-5-mini	$0.02

If you want to…	use	on model	costs about
compare models meaningfully	**tau2** (it discriminates; gsm8k saturates at ~1.0)	claude-sonnet-5	$1.62 / 10 tasks
stress long contexts, long tasks, timeouts	**appworld**	gemini-2.5-pro	$2.10–2.25 / 5 tasks
get a fast signal that nothing regressed	**gsm8k** — if it fails, stop and fix infrastructure	gpt-5-mini	< $0.01

What one task costs — and on which model. Pooled over the 279 task rows of the published v1.36 pair (`docs/results/v1.36-2026-10-12/`); `latency` is the median, tokens are the mean. **Latency was measured on a slow day for the gateway:** the median model call took 2.0× as long as in the v1.35 pair on gsm8k, 3.0× on tau2 and 1.2× on appworld (both platforms pooled), while the Service's own time per task was unchanged. Read the latency rows as that day's, and the token and dollar rows as stable. **Read the model row first:** the dollar rows are a product of tokens *and* a rate, the three benchmarks do not run the same model, and each rung's default is dearer than the one below it — so no dollar figure here transfers to a different model.

per task	gsm8k	tau2	appworld
model (the benchmark's default)	**`Azure/gpt-5-mini`**	**`aws/claude-sonnet-5`**	**`gemini-2.5-pro`**
its rate, in / out $ per 1M	0.25 / 2.00	1.52 / 7.60	1.25 / 10.00
LLM calls	1.11	11.18	28.02
input tokens	342	89,122	268,880
output tokens	177	2,078	26,382
total tokens	**519**	**91,201**	**295,261**
× a gsm8k task, in **tokens**	1×	176×	569×
cost at our gateway's rates	**$0.00054**	**$0.151**	**$0.600**
× a gsm8k task, in **dollars**	1×	279×	1,108×
cost of 100 tasks	$0.05	$15.13	$59.99
input share of tokens	66%	98%	91%
input share of **cost**	31%	90%	56%
median task latency	5.7 s	121 s	298 s
…of it inside model calls	77%	64%	95%
pass rate	0.98	0.83	0.00
tokens per **passed** task	528	~109 K	no finite value

The three benchmarks are a deliberate **difficulty ladder** — gsm8k, then tau2, then appworld, each rung roughly an order of magnitude more of everything than the one below it. **Money amplifies that ladder rather than tracking it**, because climbing a rung also switches you to a dearer model: tau2 is 176× a gsm8k task in tokens but **279×** in dollars, appworld 569× but **1,108×**. Any budget scaled off the token ratios is short by 1.6–1.9×. And **the cost share, not the token share, names the cost driver** — output is priced 4–8× input everywhere, so input is 66% of gsm8k's tokens but only 31% of its bill. Only tau2 is genuinely input-

dominated in money (90%), and it is the one benchmark where a cheaper-input model really does beat a terser one.

gsm8k's column pools its two models, which the figures below separate: 162 of its 172 rows ran gpt-5-mini and 10 ran gpt-4.1, whose per-task cost is 4.6× higher. Every one of the 279 rows is priced — no task died before its first model call — so the ratio rows divide by exactly the 519-token gsm8k task shown.

Half of appworld's LLM calls buy tool selection, not progress. The agent image (`exgentic-a2a-tool_calling`) defaults to `enable_tool_shortlisting = True, max_selected_tools = 30`: above 30 exposed tools, each turn opens with an extra LLM call that ranks the tool *names*, and only the winners' schemas reach the real call. appworld is the only benchmark in the matrix above that threshold, so its 28.02 LLM calls per task are ~14 assistant turns each preceded by a selection call — and the selection call is the dearer half, carrying every name and description against the assistant call's 30 schemas:

benchmark	LLM calls / task	of which are tool-selection calls	input tokens spent selecting	output tokens spent selecting
gsm8k	1.1	0.0 (0%)	0%	0%
tau2	11.2	0.0 (0%)	0%	0%
appworld	28.0	14.0 (50%)	67%	77%

Measured over the **279 task rows that carry chat spans** in the two matrices, both platforms. `gsm8k` and `tau2` expose fewer than the agent's `max_selected_tools = 30` and so measure **exactly zero** selection calls — they are the control for the detector. `appworld` is above the threshold and pairs **1:1**: 658 selection calls against 658 assistant calls, every turn. A selection call averages **12,954 input tokens** against **6,252** for the assistant call it precedes — it carries every tool name and description, while the assistant call carries only the 30 winners' schemas. The split is stable across clusters: 68% on OpenShift (ykt3, single-cluster) and 67% on KinD (single-node local).

So appworld's `$0.600` per task is mostly a *selection* bill, and the model never sees more than 30 of its tools at once — with `api_docs` dropped from the MCP surface, it cannot go looking for the rest (§4.4). `report.ndjson` does not separate the two call kinds; only `span_report.*` does, which is what the generator above reads. Turning it off is an agent-side setting we do not currently pass, and no matrix has measured the difference.

The rate card. Read off the LiteLLM admin UI's per-model pages on **2026-09-17** and kept in `reference/model_prices.json`, which is the single source every figure here is computed from — no generator hardcodes a price. **These are our gateway's posted rates, not an invoice and not a vendor's list price.**

model	provider	in $/1M	out $/1M	out/in	used by
`Azure/gpt-5-mini-2025-08-07`	azure	0.25	2.00	8.0×	gsm8k default
`gemini-2.5-pro`	vertex_ai	1.25	10.00	8.0×	appworld
`aws/claude-sonnet-5`	bedrock	1.52	7.60	5.0×	tau2
`Azure/gpt-4.1`	azure	2.00	8.00	4.0×	leg #4, **and the IBAC judge**

These are the ETE-Ext gateway's rates. A second gateway gets its own card under `gateways.<label>` in the same file, and its rows are priced from that card alone (§6.8). cb's two models are self-hosted (`hosted_vllm`)

with no per-token price posted, so they are carded at $0.00.

```
# regenerate every dollar figure in this section rather than editing one
python3 reference/gen-cost-analysis.py /tmp/autobench/run12-{ykt3,kind}-v136-20261012.json
```

What one leg costs. Measured totals for the canonical legs (§7.4), both platforms:

leg	tokens (OpenShift)	$ (OpenShift)	tokens (KinD)	$ (KinD)
#1 gsm8k, 1 task	406	$0.0003	470	$0.0004
#2 gsm8k, 10 tasks	5.2 K	$0.0049	4.8 K	$0.0040
#3 gsm8k, 50 tasks p=4	24 K	$0.0202	25 K	$0.0223
#9 tau2, 10 tasks	0.97 M	$1.62	0.99 M	$1.64
#10 tau2, 20 tasks p=4	1.76 M	$2.92	1.75 M	$2.89
#11 appworld, 5 tasks	1.10 M	$2.24	1.01 M	$2.11
#12 appworld, 20 tasks p=4	5.86 M	$11.83	5.91 M	$12.02
all 12 legs	**9.7 M**	**$18.67**	**9.7 M**	**$18.70**

Budget by benchmark, not by task count. All eight gsm8k legs together are **0.3%** and 0.2% of the matrix's bill; appworld's two legs are **75%** on both platforms; tau2's two are 24%. A 50-task gsm8k leg costs 2 cents — about a thirtieth of a single appworld task. The two matrices land within 0.2% of each other ($18.67 and $18.70), and #12 within 2% — but that is the same work measured twice, not a law: appworld turn counts are nondeterministic (OpenShift completed 20 appworld tasks, KinD 21), and a cluster whose tasks run longer before the 600 s per-task timeout bills more for the same request bodies. Size appworld against your own cluster.

Model choice is a cost decision too. Legs #4 and #5 ran the **identical five gsm8k tasks** at `p=4`, differing only in model:

same 5 gsm8k tasks	gpt-4.1	gpt-5-mini
pass rate (OpenShift / KinD)	0.80 / 0.80	1.00 / 1.00
LLM calls per task	2.6 – 3.2	1.0
input tokens per task	707 – 881	313
output tokens per task	58 – 60	163 – 214
total tokens per task	766 – 941	476 – 527
median task latency	11.5 – 12.8 s	8.5 – 9.2 s
cost per task	**$0.0019 – 0.0022**	**$0.00040 – 0.00051**

The reasoning model answers in **one** call; gpt-4.1 takes ~3 tool round-trips, so it sends 2.5× the input and emits about a third of the output. **gpt-4.1 costs 4.0× gpt-5-mini on this identical work** — mean of its 2 legs against gpt-5-mini's 8, the ratio spanning 2.1–5.9× depending on which pair of legs you compare, because gpt-5-mini's output length swings with reasoning effort. On our card the input side decides it alone: 2.5× the tokens at 8× the price is a **20×** input bill, and the output side cannot offset it (gpt-5-mini emits 3.1× more output at a quarter the rate, so the two output bills land within 9% of each other).

Which model is cheaper is a property of your price list, not of the models. Compute it rather than inheriting our answer:

```
cost per task  =  (input_tokens × P_in  +  output_tokens × P_out) / 1e6
```

Every number above is a floor, for five reasons. A task killed by `task_timeout_seconds` burns tokens but leaves no `report.ndjson` row, so the three appworld tasks that timed out are missing from these totals. tau2's user simulator runs in the **MCP** pod, which is not instrumented — its inference is billed by the gateway and counted nowhere here (the tell: 23% of a tau2 task's wall time sits inside tool calls, against 7% for gsm8k and under 1% for appworld, and every `chat` span carries the *agent's* model). A leg that replays the gateway's completion cache re-reports stored `usage` for calls that were never made upstream, so a cache-contaminated leg can also read *high*. Plugins add judge calls that are billed but not in the agent's spans: the IBAC judge runs `Azure/gpt-4.1` on a fixed 1,577-char system prompt, so **≥ $0.00111 per authorized tool call** — 2.5× the whole gsm8k task it guards, and 1.2–2.9× the agent's own bill across legs #6–#8 (see `PLUGIN_OVERHEAD.md`). Conversely the *input* figures are an upper bound: the gateway reports `usage.prompt_tokens_details.cached_tokens` but publishes no cached-input rate, and `report.ndjson` stores one undifferentiated `llm_input_tokens`, so no past run can be re-priced. If cached input were free, tau2 would floor at $0.0158 a task and appworld at $0.264 — the `input share of cost` row is the bound.

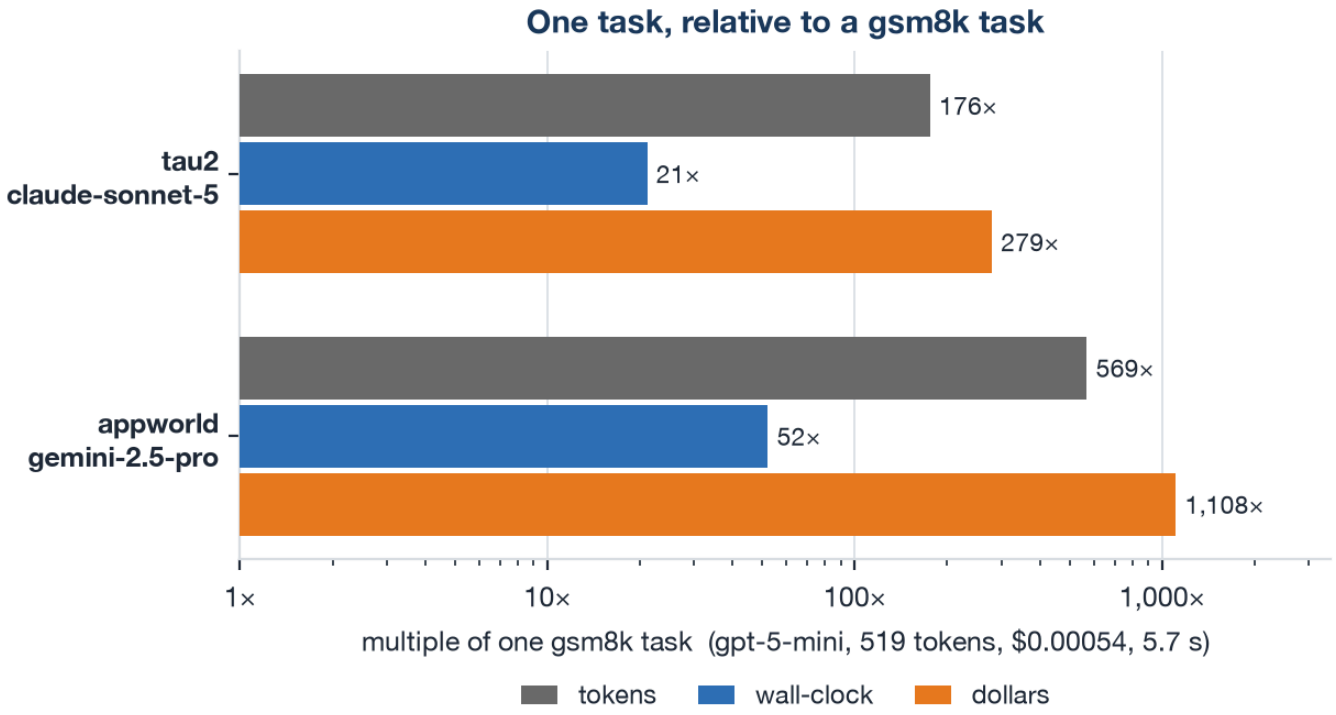
Token- and cost-efficiency in six figures

Six things we can now state as measurements rather than intuitions. Each figure carries **one** take-away, in bold under it; the sentence and the picture are produced together by one generator, so they cannot drift apart:

```
uv run --with matplotlib python reference/gen-cost-charts.py \
    /tmp/autobench/run12-ykt3-v136-20261012.json /tmp/autobench/run12-kind-v136-20261012.json
```

That writes `docs/img/*.png`, the take-away text it embeds into this section, and `docs/img/takeaways.json` — which is what the slide deck reads, so the deck's captions are the same strings. Nothing below is hand-written: to change a number, change the artifacts or `reference/model_prices.json` and re-run.

Figure 1 — Money climbs the ladder faster than tokens

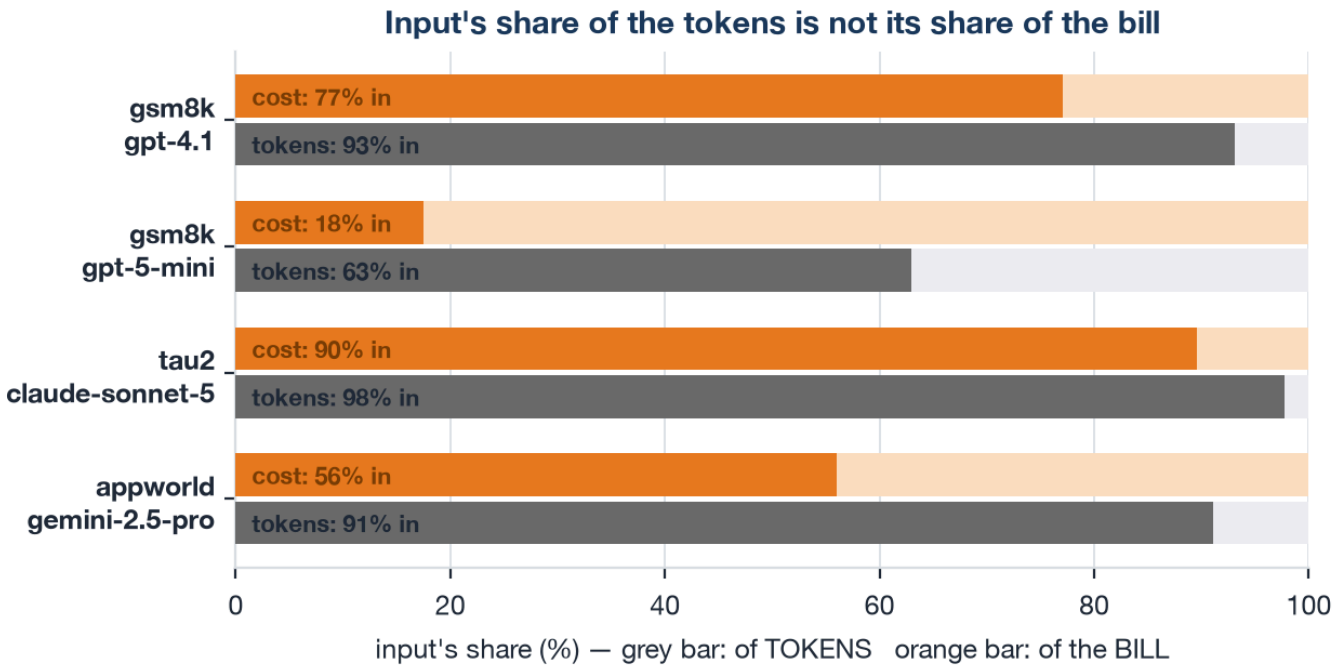


Money climbs the difficulty ladder faster than tokens do: tau2 is 176× a gsm8k task in tokens but 279× in dollars, appworld is 569× a gsm8k task in tokens but 1,108× in dollars — so a budget scaled off the token ratios is short by 1.6–1.9×.

Each rung is a different model, which is why the dollar bar outruns the token bar: climbing the ladder also buys a dearer model. Wall-clock climbs SLOWEST of the three (21× tau2, 52× appworld), because the harder benchmarks parallelize their turns while their bills add up.

gsm8k baseline: 519 tokens, $0.00054, 5.7 s median, over 172 task rows.

Figure 2 — Read the share of the BILL, not the share of tokens

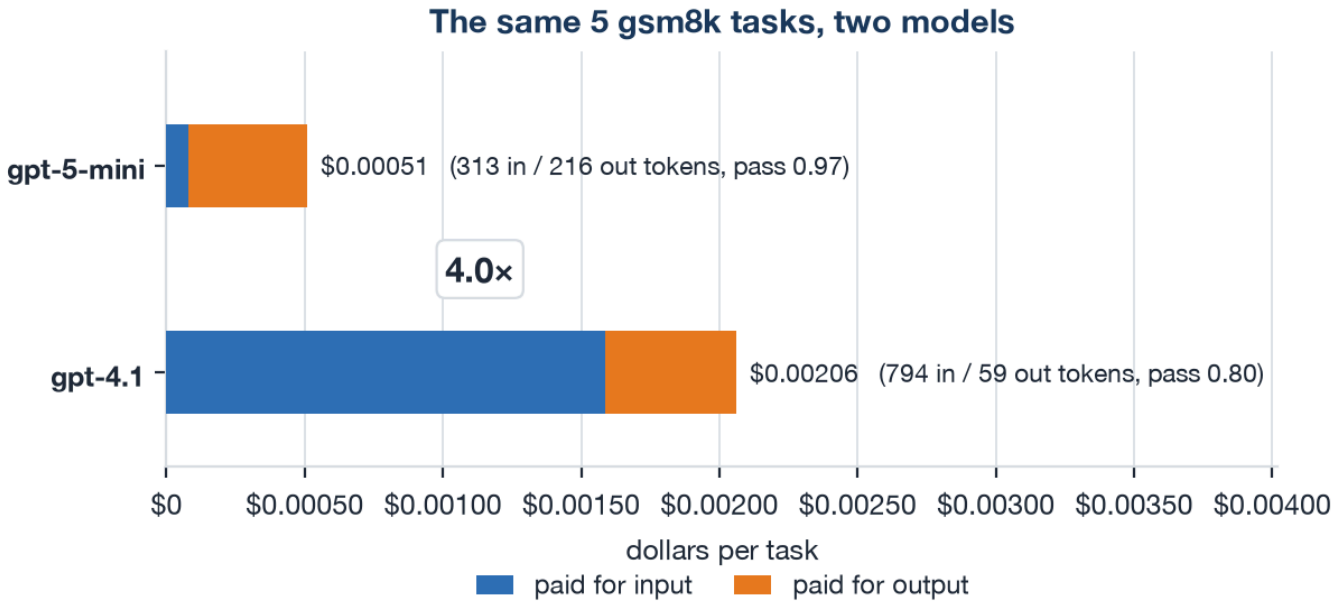


Output is priced 4–8× input on every model we use, so the token split misstates the bill every time: gsm8k on gpt-5-mini is 63% input by tokens and only 18% by cost.

The gap is the reason a model can win on token count and lose on the invoice. It also says where to look for savings: only tau2 is genuinely input-dominated in money, so it is the one benchmark where a cheaper-input model beats a terser one — everywhere else verbosity is the thing to control.

Both bars are per-task means over every priced row of that benchmark and model.

Figure 3 — Model choice is settled on the input side

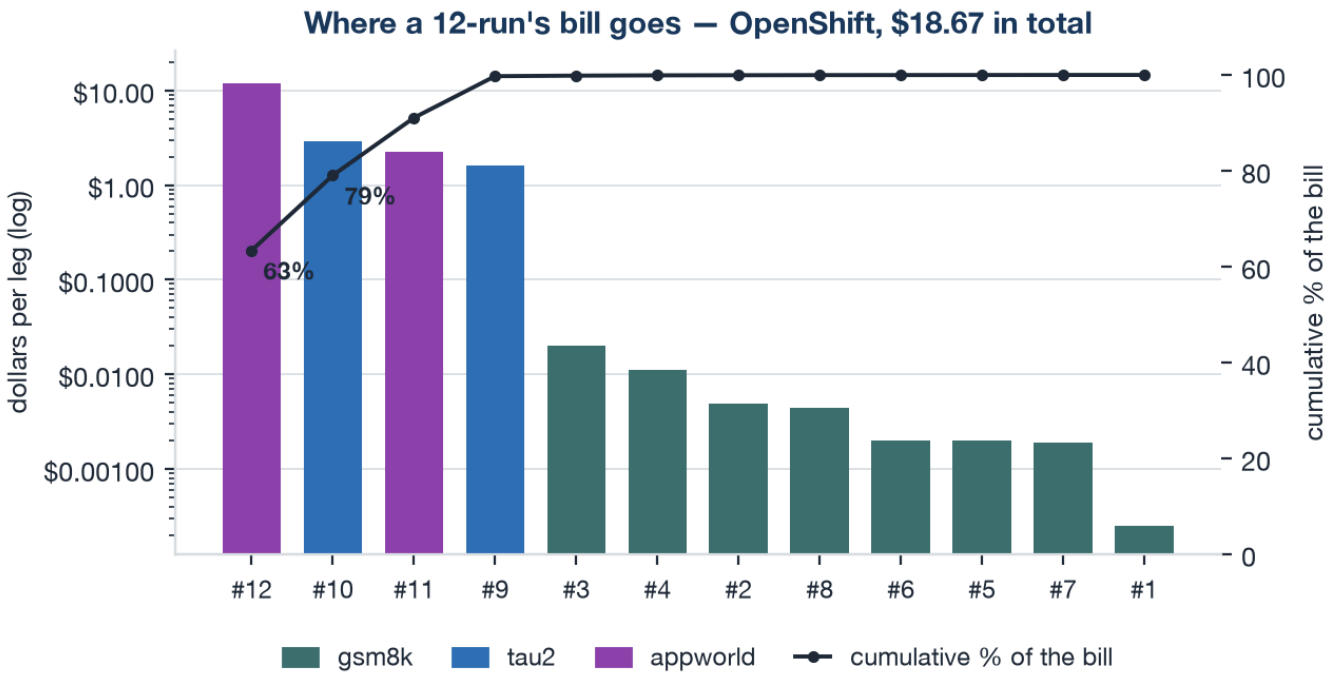


gpt-4.1 costs 4.0× gpt-5-mini on the identical 5 gsm8k tasks, and the input side decides it alone: 20× the input bill, against an output bill the two models split within 9%.

Mean of gpt-4.1's 2 legs against gpt-5-mini's 8; leg to leg the ratio spans 2.1×–5.9×, because the reasoning model's output length swings with how much it reasons. Note the shape rather than the winner: gpt-5-mini answers in one call and emits MORE output, gpt-4.1 takes ~3 tool round-trips and sends far more input. Which one is cheaper is a property of the price list, not of the models.

The task set is identical and task selection is deterministic, so this is the one comparison in the matrix with no workload difference to explain away.

Figure 4 — Two legs of twelve are most of the bill

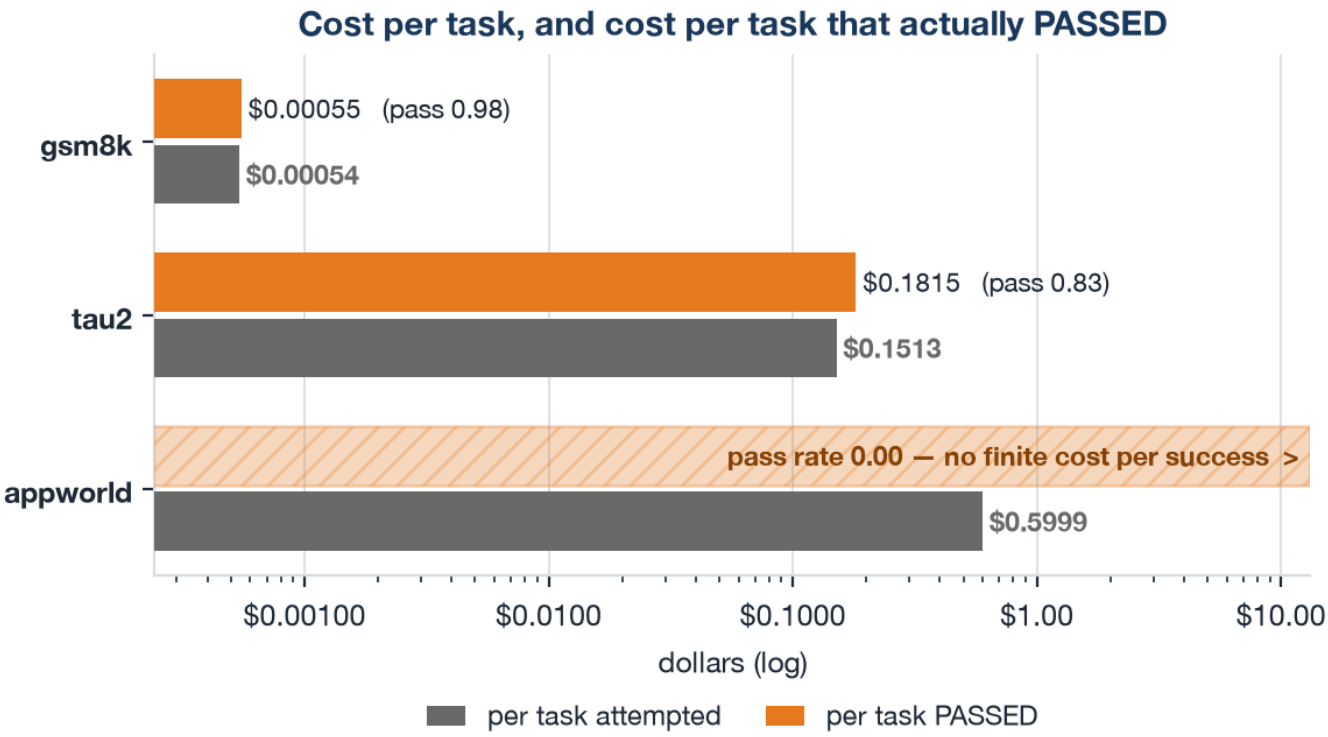


Two of the twelve legs are 79% of a matrix's bill, while all 8 gsm8k legs together are 0.3% of it — so budget by benchmark, not by task count.

The bars are log-scaled because a linear axis renders every gsm8k leg as nothing at all, which is itself the finding: there are only two legs worth watching. It is also why the cheap legs are the ones to iterate on — a full gsm8k sweep costs less than a rounding error on one appworld leg.

OpenShift; the ordering reproduces on the other platform, the absolute appworld figures do not (nondeterministic turn counts).

Figure 5 — Efficiency only means anything per SUCCESS

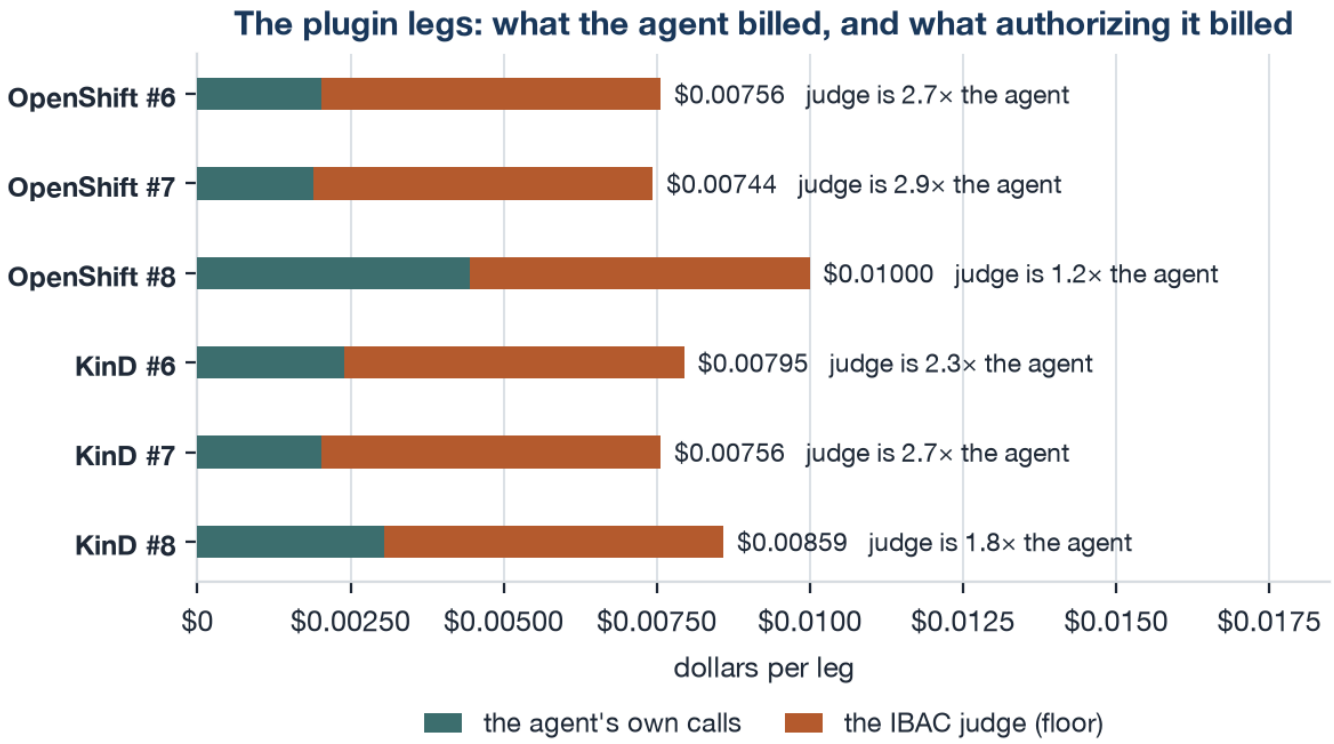


Dividing by the pass rate is what turns a token count into an efficiency figure: gsm8k $0.00054 per task becomes $0.00055 per PASS; tau2 $0.1513 per task becomes $0.1815 per PASS — and appworld has no finite cost per success at all, because it passed nothing.

A change that halves your token use and halves your pass rate has gained you nothing, which is why we never quote tokens per task as an efficiency number on its own. The unbounded row is not a rendering artifact: it is the honest way to report a benchmark that runs to completion, records every token, and then fails evaluation.

Pass rates here are over PRICED rows; the headline rates in the reports use the tasks attempted, which is a lower number for appworld because a timed-out task leaves no row.

Figure 6 — On the plugin legs, the check outcosts the work



One IBAC judge completion costs at least $0.00111 — 2.5× the entire gsm8k task on gpt-5-mini it is authorizing — so across legs #6/#7/#8 the judge bills 1.2–2.9× what the agent does.

The judge runs gpt-4.1, the dearest model on our card, against a FIXED 1,577-character system prompt that does not shrink with the task — so the cheaper the work, the more lopsided this gets. None of it appears in report.ndjson: the sidecar makes the call, not the instrumented agent, which is why every other figure in this section is a FLOOR.

Judge cost is tool calls × the floor per call; the proposed-action block that follows the fixed prompt is not counted. tau2's user simulator is invisible the same way, inside the uninstrumented MCP pod.

The three benchmarks and the 12 runs, compared

The figures above price the work. This is what the work *is*, and what having run it is worth.

The three benchmarks. They differ in the shape of a task, not just its size — which is why the ladder is a ladder and not three sizes of the same test:

	gsm8k	tau2	appworld
A task is	one grade-school word problem, answered in text	one customer-service conversation in the `retail` domain, against a simulated user	one multi-app scenario (email, phone, shopping…) automated through an API surface
Turn structure	one model call, ~1 tool call	~11 model calls alternating with a **user simulator** that replies in character	~30 model calls over a long horizon, every call re-sending the whole conversation
Ends when	the answer is emitted	the dialogue reaches a resolution or the policy is violated	the scenario's goal state is reached, or the 600 s task timeout kills it

	gsm8k	tau2	appworld
Scored by	exact numeric match	task-completion + policy compliance, per tau2's own scorer	appworld's own state assertions — all-or-nothing
It exists to test	that the *plumbing* works: deploy, auth, telemetry, scoring, S3 export	that the agent can **hold state across turns** and use tools under a policy	that the agent survives **long horizons** — context growth, timeouts, partial failure
A result is worth	a go/no-go on infrastructure. It saturates near 1.0, so it cannot rank models	a genuine model/configuration comparison — it discriminates, and its 0.90 leaves headroom in both directions	a stress signal, not a capability score: at 0.00 it tells you what *breaks*, not who is better
Watch out for	a pass rate of 1.0 proves nothing about the agent	its user simulator's inference is billed but **not** in our telemetry	long tasks can hit the 600 s timeout (1 of 50 here), and a killed task leaves no `report.ndjson` row at all

Where their value actually lies. gsm8k is the only one cheap enough to run on every change, and that is its entire point — it is a **smoke test with a score**, and treating its 0.98 as a model measurement is the most common misreading of these numbers. tau2 is the only rung that discriminates: it is multi-turn, so it fails in *informative* ways (a wrong answer, a policy violation and a dropped thread are different failures), and its pass rate moved with configuration in our matrices while gsm8k's did not. appworld earns its place precisely because it fails: a benchmark that nothing passes still answers "does the platform survive a 4-minute, 300 K-token task at 4-way concurrency?", and it is the only leg that has ever exposed a timeout, a context limit or a cache effect before a user did.

The 12 runs. One matrix is 12 legs, each changing exactly one thing against the leg before it, so a difference has one candidate explanation:

legs	what they parameterize	what having run them established
#1–#3 gsm8k 1 → 10 → 50 tasks, p=1 → 4	volume, then concurrency	the pipeline is stable and **deterministic**: 7 of 12 legs have byte-identical input-token totals across two clusters, which is the strongest like-for-like check available
#4 gsm8k on gpt-4.1	model swap on **identical** tasks	the only clean model comparison in the matrix — `gpt-4.1` costs **4.6×** `gpt-5-mini` for no pass-rate gain at this difficulty
#5–#8 gsm8k under AuthBridge: auth-only, ibac-only, full, full + per-plugin override	one security layer at a time, same five tasks	plugin cost lands in the **IBAC judge** on both clusters: +1.36 s/task on OpenShift, +1.48 s on KinD; the sidecar's presence and every other layer are below the noise floor (`PLUGIN_OVERHEAD.md`)
#9–#10 tau2 10 → 20 tasks, p=1 → 4	multi-turn, then multi-turn under load	multi-turn works end to end, including the user simulator — and tau2 is where pass rates carry information (0.85–1.00 across sides)
#11–#12 appworld 5 → 20 tasks, p=1 → 4	long horizon, then long horizon under load	the limits are real and they are **upstream**: a 0.00 pass rate, the matrix's only timeout, and ~77% of its bill in two legs

What the matrix as a whole is worth. Its value is not the pass rates — it is that **the same 12 request bodies produce comparable measurements on two unlike clusters**. In the published v1.36 pair: token capture was complete (0 of 279 rows lost their usage span), every task reached the model (0 health-probe losses), nothing was lost to infrastructure, and 9 of 12 pass rates matched exactly. The 3 that differed are each one task — one task moves a 5-task leg by 0.20 — which is why the reports carry a **per-cause** failure table

(timeout 0/1, upstream-agent defect 4/4, wrong answer 1/1): a task lost to a socket and a task lost to a wrong answer land in the same denominator and only one of them says anything about the agent. Read the cause table, then the rate.

Two limits of the matrix worth stating in the same breath. **Absolute latency does not travel**: the same condition on the same image measured up to 1.7× apart between our clusters, so seconds are always qualified by cluster while pass rates and token counts travel fine. And **the deploy, not the task, is the unit of replication** — every task in a leg shares one deployment, so adding tasks tightens the wrong interval; if you need a tighter number, add deploys.

Newcomer-facing versions of these tables, with what each benchmark actually is: `BENCHMARKS_PRIMER.md`.

---

3. Getting a caller token

The caller JWT is a normal Keycloak token from the instance's realm. Obtain it via the password grant (the same Direct-Access-Grants flow the dev/test users use). The realm and token endpoint are derived from the instance `iss` (`<iss-origin>/realms/<realm>/protocol/openid-connect/token`).

```
# --- Environment (pick the block for your target cluster) ---
# Option A — ykt3 (OpenShift; realm/client "rossoctl"):
export SVC="https://autobench-rossoctl-system.apps.ykt3.example.com"   # AutoBench Service base
   URL
export KC="https://keycloak-keycloak.apps.ykt3.example.com"            # Keycloak base (iss
   origin)
export REALM="rossoctl"                                                # realm from the iss
export KC_CLIENT="rossoctl"                                            # public client with
   Direct Access Grants
# Option B — kind-rossoctl (local; realm/client "rossoctl"):
#   The instance's iss is http://keycloak.localtest.me:8080/realms/rossoctl, but in-cluster the
#   Service reaches Keycloak via keycloak_backchannel_url. You still fetch YOUR token from the
#   externally reachable route below (localtest.me resolves to 127.0.0.1).
# export SVC="http://autobench.localtest.me:8080"
# export KC="http://keycloak.localtest.me:8080"
# export REALM="rossoctl"
# export KC_CLIENT="rossoctl"
# Attribution matters: config endpoints require preferred_username == "benchmarker".
# Any valid realm user works for deploy/run/report.
# Never hardcode the password — supply it out-of-band (env, secret manager, or an
# interactive prompt). e.g.: read -rs -p "benchmarker password: " KC_PASS; export KC_PASS
export KC_USER="benchmarker"
export KC_PASS="${KC_PASS:?set KC_PASS in your environment; do not commit it}"
export TOKEN=$(curl -s "$KC/realms/$REALM/protocol/openid-connect/token" \
  -d grant_type=password \
  -d client_id="$KC_CLIENT" \
  -d username="$KC_USER" \
  -d password="$KC_PASS" \
  | python3 -c 'import sys,json; print(json.load(sys.stdin)["access_token"])')
# Sanity check: the Service echoes the validated claims + which instance you routed to.
curl -s "$SVC/hello" -H "Authorization: Bearer $TOKEN" | python3 -m json.tool
```

`GET /hello` returns `{iss, preferred_username, rossoctl_base_url, claims}`. If you get `401`, the token is missing/invalid; `403 issuer not in scope` means no instance file matches the token's `iss` (see §4.1).

> **Why the kind block differs:** in-cluster the public `iss` host is unreachable, so the instance file sets `keycloak_backchannel_url` and the Service dials *that* for JWKS/ROPC. This only affects the Service's own dialing — your caller token still comes from the externally reachable Keycloak route (Option B above).

---

4. Onboarding a benchmark (one-time, per cluster/instance)

These steps are **out-of-band** — the Service verifies them but cannot perform them.

4.1 Instance config file (`instances/<encoded-iss-host>.json`)

One file per instance keys the whole thing to the `iss`. Loaded at startup from `settings.instances_dir`. Shape:

```
{
  "iss": "https://keycloak-keycloak.apps.ykt3.example.com/realms/rossoctl",
  "keycloak_backchannel_url": null,
  "rossoctl_base_url": "http://rossoctl-backend.rossoctl-system:8000",
  "service_credential": {
    "client_id": "rossoctl",
    "client_secret": "",
    "username": "benchmarker",
    "password": "<benchmarker-password>"
  },
  "mlflow": { "tracking_url": null },
  "s3": { "bucket": null },
  "mcp_endpoint_template": null,
  "agent_endpoint_template": null,
  "workload_otel": null
}
```

- `service_credential` is the ROPC identity the Service presents to Rossoctl. It must exist in the realm with whatever profile fields the realm requires (`firstName` / `lastName` , and on some realms `email`), or Keycloak 400s with "Account is not fully set up"; `reference/keycloak-ensure-user.sh` fills them where unset (Admin Guide §3.6).
- `mcp_endpoint_template` / `agent_endpoint_template` stay `null` when the workloads run on the Service's own cluster, as in every example here: the Service reaches them over in-cluster service DNS. When they run on another cluster, set both to that cluster's Route shape, `https://{service}-{namespace}.apps.<cluster>.example.com` (Admin Guide, §5).
- `mlflow` / `s3` can be seeded here or set later via `PUT /config` (§5).

4.2 Infrastructure resources (baked into the benchmark definitions)

The Service sends CPU/memory requests+limits in the create body, so no post-create patch is needed. Current values (from `benchmarks/registry.py`):

Workload	requests	limits
MCP tool	`500m` CPU / `512Mi`	4 CPU / `4Gi`
A2A agent	`500m` CPU / `512Mi`	4 CPU / `2Gi`

4.3 Workload secrets (provisioned out-of-band as cluster Secrets)

The Service has **no Secrets API**. It references these by name in the deploy body; if a Secret is missing the workload never becomes Ready, and the run precheck returns `424` naming exactly what to provision.

Benchmark	Secret (name → key)	Consumed by	Purpose
gsm8k	`hf-secret` → `hf-token`	tool	HuggingFace dataset access
gsm8k / tau2 / appworld	`openai-secret` → `apikey`	agent (+ the tau2 tool, for its user simulator)	LiteLLM API key

Provision them on the workload cluster before deploying, e.g.:

```
kubectl -n team1 create secret generic hf-secret --from-literal=hf-token="$HF_TOKEN"
kubectl -n team1 create secret generic openai-secret --from-literal=apikey="$LITELLM_KEY"
```

> `openai-secret` **must exist before you run the Service — provision it out-of-band.** On clusters where the rossoctl platform chart owns the agent namespaces (e.g. `team1`), the chart only manages `openai-secret` when its Helm value `secrets.openaiApiKey` is set. Leave that value **empty** and create the Secret yourself (command above / a secret manager / External Secrets), so the credential lives outside Helm values and a later `helm upgrade` **cannot** overwrite it with an empty key. When the value is empty the chart skips the Secret entirely and its install NOTES print a `WARNING: secrets.openaiApiKey is NOT set` reminder. If you instead let the chart template the key, every `helm upgrade` re-applies whatever is in the release values — an empty value there silently zeroes the key mid-flight and every LLM call fails until the Secret is restored and the workload pods are `rollout restart`ed (pods read `apikey` via `secretKeyRef` only at startup).

4.4 What each benchmark bakes in, and what it needs from you

The division of labour is easy to get backwards: **the benchmark itself — dataset, world, evaluator — is inside the MCP image**, and `tool_env` (`registry.py`) adds only the credentials and the per-benchmark quirk overrides. There is no config knob for the dataset or the domain.

	what's baked in	`tool_env` it needs
gsm8k (`exgentic-mcp-gsm8k`)	the HuggingFace dataset loader, pinned to the `main/test` split — those 1,319 rows are fetched at pod startup	`HF_TOKEN` (from `hf-secret`), plus `EXGENTIC_SET_BENCHMARK_RUNNER=direct`
tau2 (`exgentic-mcp-tau2`)	the τ²-bench library + its `retail` domain (114 tasks), and a user-simulator LLM	`OPENAI_API_KEY` (from `openai-secret`) + `EXGENTIC_SET_BENCHMARK_ACTION_TIMEOUT=1000` — it makes its own inference calls (flow 4)
appworld (`exgentic-mcp-appworld`)	the whole app-suite sandbox (`exgentic install --benchmark appworld`) — upstream appworld at commit `edc96012` wrapped in a *custom* tool-per-API adapter, served from the `test_normal` split (168 tasks)	just `BENCHMARK_NAME` — upstream's `.env.appworld` is explicitly empty

Every benchmark also gets `BENCHMARK_NAME` and an `OPENAI_API_BASE` that the Service **injects per deploy** from the instance's `workload_llm.api_base` (§4.1) — never baked into the image; a deploy with no gateway

configured is rejected with `422` rather than falling back to a default. tau2's simulator model is injected the same way, from the run's model.

Two edges worth knowing before you copy env between benchmarks:

- **appworld rejects the action-timeout override tau2 requires** and crashes at startup with `Unknown benchmark override 'action_timeout'`. The env is per-benchmark, not a shared default.
- **`hf-secret` must exist for gsm8k even though the dataset is public.** Without it the MCP pod sits in `CreateContainerConfigError` and the agent crash-loops; with an empty value it works.

The **agent** side, by contrast, is the same everywhere: one image (`exgentic-a2a-tool_calling:latest`) is the only entry in all three `agents={…}` maps, gaining just a `-<benchmark>` name suffix. So the benchmark lives in the MCP pod and the subject under test is the same binary every time.

4.5 MLflow + OTEL collector (optional, for reports)

Reporting is fail-soft: if MLflow client-creds aren't configured, runs still succeed and export to S3, but `report.ndjson` is empty and the report endpoints return `409`. To enable, provision an OTEL collector **on the workload cluster** (it forwards the agents' spans to the MLflow beside the Service) out-of-band, and set the MLflow read creds via `PUT /config`.

---

5. Instance-specific Service config (`/config`)

`GET` / `PUT /config` are **benchmarker-only** (`preferred_username == "benchmarker"`; else `403`). They set only what the Service itself enacts — MLflow (read side) and S3. Attempting to set a workload credential is rejected with **`422`** (`extra="forbid"`). Secrets are redacted in responses.

```
# View effective config for your instance (file defaults + any overrides), secrets redacted.
curl -s "$SVC/config" -H "Authorization: Bearer $TOKEN" | python3 -m json.tool
# Point the Service's result sink at an S3 bucket + wire up MLflow read creds.
curl -s -X PUT "$SVC/config" \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d '{
        "s3": {
          "bucket": "rossoctl-benchmarking",
          "region": "us-east-1",
          "access_key_id": "AKIA...REDACTED",
          "secret_access_key": "REDACTED",
          "public_read": false
        },
        "mlflow": {
          "tracking_url": "https://mlflow.redhat-ods-applications.svc.cluster.local:8443",
          "client_id": "mlflow-client",
          "client_secret": "REDACTED",
          "token_url": "https://keycloak-
    keycloak.apps.ykt3.example.com/realms/rossoctl/protocol/openid-connect/token",
          "insecure_tls": false
        }
      }'
```

Rejected example (workload cred → `422`):

```
curl -s -o /dev/null -w '%{http_code}\n' -X PUT "$SVC/config" \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d '{"hf-secret": "..."}'          # -> 422
```

A `PUT` is a **field-level merge**: fields you do not mention keep the value they had, from the instance file or from an earlier `PUT`. Whole sections behave the same way — a body carrying only `mlflow` leaves `s3` untouched.

Because unmentioned and `null` mean different things, an explicit `null` **clears** a field — the only way to unset one at runtime:

```
curl -s -X PUT "$SVC/config" \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d '{"mlflow": {"workspace": null}}'     # stop sending x-mlflow-workspace
```

5.1 `GET /mlflow/health` — is the MLflow you just configured actually usable?

Same auth as `/config`, and the natural follow-up to it: a `PUT` that returns 200 proves the config parsed, not that MLflow accepts it. This probes the effective config in four independent stages — `auth`, `read`, `write`, `round_trip` — and **always answers 200**, because "the probe could not run" and "MLflow is broken" are different findings and an HTTP status cannot carry both.

```
curl -s "$SVC/mlflow/health" -H "Authorization: Bearer $TOKEN" | python3 -m json.tool
curl -s "$SVC/mlflow/health?round_trip=false" -H "Authorization: Bearer $TOKEN"   # read stages
   only
```

```
{
  "ok": true,
  "tracking_url": "https://mlflow.redhat-ods-applications.svc.cluster.local:8443",
  "credential_mode": "bearer_token",
  "experiment_id": "1",
  "workspace": "team1",
  "insecure_tls": true,
  "auth": {"ok": true, "detail": "bearer obtained via bearer_token"},
  "read": {"ok": true, "detail": "traces API answered; experiment 1 is non-empty"},
  "write": {"ok": true, "detail": "probe spans exported"},
  "round_trip": {"ok": true, "detail": "probe trace readable after 2 attempt(s)"}
}
```

`write` and `round_trip` are the reason this exists, and they are why it is not a read-only check. The Service emits four spans of its own per run, and a trace whose root `Agent.Session` span never arrives is **dropped** when the report is assembled — so a broken export yields `pass_rate 1.0` beside a zero-byte `report.ndjson`, with no error anywhere. The read path answers 200 throughout that, since reads go through `httpx` honouring `insecure_tls` while the span exporter is a separate library needing a real trust anchor and an RBAC grant. `write` therefore emits one synthetic trace through the real exporter and captures the `opentelemetry` logger for its duration, because `BatchSpanProcessor` exports on a worker thread and **logs** failures rather than raising — so the cause comes back in `write.error` instead of sitting in a pod log. The capture watches `WARNING` as well as `ERROR`, which is load-bearing rather than defensive: the exporter splits a *retryable* failure across both levels, putting the cause in a per-attempt `Transient error … retrying in 0.96s.` warning and leaving the terminal `ERROR` generic (`Failed to export span batch due to timeout, max retries or shutdown.`), while a *non-retryable* one logs a single self-describing `ERROR` (`Failed to`

`export span batch code: 403, reason: Forbidden`). An ERROR-only handler therefore names the cause for the 403 and never names it for the missing TLS anchor. Both levels are read and spliced, and because a warning alone means an attempt failed and the *retry succeeded*, only an `ERROR` marks the stage failed; a recovered retry is reported in `detail`. `round_trip` then reads that trace back, which is the only way to catch an MLflow that accepts the POST and drops it.

The response is a **whitelist** of fields, the same discipline `span_report.*` follows: the probe holds a bearer and possibly a password, and `credential_mode` names the shape without disclosing the material. Fields are absent, not false, for a stage that never ran — the handler stops after a failure instead of piling a second cause onto it.

Two properties make it safe against a live install: the probe **cannot pollute a report**, because every report is filtered to its own run's session ids and the probe's is a fresh `uuid4`, and it costs **no LLM gateway call**. It does write one trace per call. `autobench-cli mlflow-health` wraps it (§7.1) and exits `8` when any stage failed, so an install script and an operator gate on the same probe.

---

6. Benchmark lifecycle

The happy path is: **deploy → wait for Ready → run → poll → report → download from S3**.

Every block below is copy-pasteable once these four variables are exported (see §3 for how to get the token; §7.1 wraps this whole section in a single command if you would rather not paste):

```
export SVC="https://autobench-rossoctl-system.apps.ykt3.example.com"   # Service base URL
export BENCH=gsm8k          # gsm8k | tau2 | appworld
export SCOPE="namespace=team1&agent=tool_calling&experiment=default"
export TOKEN="…"            # from §3; never echo this
# Self-signed OpenShift route? add -k to every curl, or: export CURL_OPTS=-k
```

6.0 Discover what's available

```
curl -s "$SVC/benchmarks" -H "Authorization: Bearer $TOKEN" | python3 -m json.tool
curl -s "$SVC/benchmarks/gsm8k" -H "Authorization: Bearer $TOKEN" | python3 -m json.tool
curl -s "$SVC/namespaces" -H "Authorization: Bearer $TOKEN" | python3 -m json.tool
```

Three benchmarks are registered: `gsm8k` (single-turn), `tau2` (multi-turn, runs a server-side user-simulator LLM), `appworld` (registered; see §8).

6.1 Deploy (create the MCP tool + A2A agent)

`POST /benchmarks/{name}/deploy` creates both the shared MCP tool (`exgentic-mcp-<name>`) and the agent (`exgentic-a2a-<agent>-<name>[-<experiment>]`). Returns `201`.

```
curl -s -X POST "$SVC/benchmarks/gsm8k/deploy" \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d '{
        "agent": "tool_calling",
        "namespace": "team1",
        "experiment": "default"
      }' | python3 -m json.tool
```

Request fields (`DeployBenchmarkRequest`):

Field	Default	Meaning
`agent`	`tool_calling`	agent flavor (only `tool_calling` today)
`model`	`null`	**deploy-time** model, baked into agent env; `null` → benchmark default (`openai/Qwen3.6-35B-A3B`)
`namespace`	`team1`	target namespace
`experiment`	`default`	non-default suffixes the agent name so variants coexist
`authbridge_enabled`	`false`	inject the AuthBridge sidecar with the **cluster-default** pipeline (layer-2 knob)
`plugin_preset`	`null`	layer-3 preset (`auth-only`|`ibac-only`|`full`); forwarded to the backend as `pluginPreset` → `AgentRuntime.spec` → operator renders the per-agent pipeline. Requires `authbridge_enabled=true`
`plugins`	`null`	per-plugin policy overrides as `["NAME:POLICY"]` tokens (`POLICY` = `enforce`|`observe`|`off`); forwarded as `plugins`. Requires `authbridge_enabled=true`
`on_error`	`null`	chain-default policy (`enforce`|`observe`|`off`); forwarded as `onError`. Requires `authbridge_enabled=true`
`plugin_config_file`	`null`	**rejected with 422** — a local filesystem path with no HTTP analog; use `plugin_preset`/`plugins`/`on_error` instead
`llm_api_base`	`null`	this deployment's LLM gateway (an `http(s)` URL), in place of the instance's `workload_llm.api_base`. Requires `llm_gateway`. From v1.36
`llm_key_secret`	`null` → `openai-secret:apikey`	`NAME` or `NAME:KEY` of a Secret in `namespace` holding that gateway's API key. The key never passes through the Service. From v1.36
`llm_gateway`	`null` → the instance's `workload_llm.gateway_label`, else `default`	the label the artifacts publish for the gateway. The URL is never published. From v1.36

From v1.36 an unknown field is a `422`, not silently ignored. The response also carries `llm: {gateway, model, key_secret}`, which is what the deployment will actually call. Check it, because an older Service answers `201` and quietly deploys on its own gateway. See §6.8.

Model swap (run #4 pattern): the deploy endpoint always (re)creates the *shared* tool, so re-deploying with a new model 409s on the existing tool. Deploy the model-swapped agent alone via `POST /agents` under a distinct experiment instead:

```
# Generate the agent body with build_agent_request semantics, then POST /agents directly.
curl -s -X POST "$SVC/agents" \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d '{
        "name": "exgentic-a2a-tool-calling-gsm8k-azmini",
        "namespace": "team1",
        "container_image": "ghcr.io/exgentic/exgentic-a2a-tool_calling",
        "image_tag": "latest",
        "env_vars": [
          {"name": "MCP_URL", "value": "http://exgentic-mcp-gsm8k-
    mcp.team1.svc.cluster.local:8000/mcp"},
```

```
          {"name": "LLM_MODEL", "value": "openai/Azure/gpt-4o-mini"},
          {"name": "EXGENTIC_SET_AGENT_MODEL", "value": "openai/Azure/gpt-4o-mini"},
          {"name": "OPENAI_API_KEY", "value_from": {"secret_key_ref": {"name": "openai-secret",
    "key": "apikey"}}}
        ],
        "service_ports": [{"name": "http", "port": 8080, "target_port": 8000}]
      }' | python3 -m json.tool
```

6.2 Wait until Ready

```
curl -s "$SVC/benchmarks/gsm8k/status?namespace=team1&agent=tool_calling&experiment=default" \
  -H "Authorization: Bearer $TOKEN" | python3 -m json.tool
```

Returns `tool_ready` / `agent_ready` booleans plus raw `readyStatus`. Poll until both are `true`. If a workload stays not-Ready, the run precheck (§6.3) will name the missing Secret.

6.3 Submit a run

`POST /benchmarks/{name}/runs` runs a **cluster-API-free precheck** (tool+agent deployed and Ready) then returns **`202`** with a `run_id`. Prechecks:

- `409` — not deployed (`POST …/deploy` first).
- `424` — deployed but not Ready; the message names required Secret(s) (e.g. `hf-secret`).

```
export RUN=$(curl -s -X POST "$SVC/benchmarks/gsm8k/runs" \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d '{
        "agent": "tool_calling",
        "namespace": "team1",
        "experiment": "default",
        "max_tasks": 10,
        "max_parallel_sessions": 1,
        "timeout_seconds": 600
      }' | python3 -c 'import sys,json; print(json.load(sys.stdin)["run_id"])')
echo "run_id=$RUN"
```

Run fields (`RunRequest`) are all **run-time** knobs:

Field	Default	Notes
`max_tasks`	1	number of benchmark tasks to evaluate; **capped by the task pool, silently** — see below. What a given count costs in tokens and minutes: §2, Picking a benchmark
`max_parallel_sessions`	1	concurrency
`timeout_seconds`	300	whole-run wall-clock ceiling; raise for large tau2 runs
`agent` / `namespace` / `experiment`	—	must match a deployed agent
`model`	`null`	**not** forwarded per-session; model is fixed at deploy time

> **What `max_tasks: N` actually means.** N **independent episodes against one deployment** — not one long session, and not N deploys. The runner fetches the task list, slices off the first N, and for each id runs the same three-step sequence — `MCP.CreateSession` → `Agent.Call` (A2A `send_prompt`) →

> `Evaluator.Evaluate` — inside its own `Agent.Session` OTEL span keyed by that `task_id` (`runner/engine.py`); One task, end to end traces those three steps together with what the agent does inside the middle one. Nothing carries over between tasks: no shared conversation, no shared MCP session, no ordering dependence. Failures are isolated per task (each `_one` has its own `asyncio.timeout` and the batch is gathered with `return_exceptions=True`), so one wedged task costs you that task, not the run — and the summary is republished after **every** task, so even a run the wall clock kills leaves usable partial results. That independence is what makes `pass_rate = evaluated_pass / total` a rate rather than an average of correlated trials.
>
> `max_tasks` is **how many**; `max_parallel_sessions` is **how many at once** (a separate semaphore). They are orthogonal: `max_tasks: 50, max_parallel_sessions: 4` is fifty problems with four in flight. The agent and MCP pods are shared by all N tasks, which is what makes a multi-task run cheap relative to its task count — and also why per-task *latency* from a short run is dominated by the first concurrency wave's warm-up.

> `max_tasks: 1` **against a 114-task pool is not an error either** — it is the same slice seen from the other side. You get `task_id 0` and only `task_id 0`; the other 113 are never created, and nothing reports them as skipped. Both directions of mismatch are silent, and `summary.total` is the one field that tells you what actually ran: your number when you under-ask, the pool size when you over-ask. The habit to build: **a 1-task run is a smoke test, not a measurement.** It proves the deploy, auth, LLM reachability and telemetry path work end to end — that is exactly why leg #1 of the canonical matrix is one gsm8k task — but it says nothing about capability, because it is a single pass/fail on the *same* problem every time. Because the slice is a prefix, every leg of a benchmark re-runs that first task: legs #1 ⊂ #2 ⊂ #3, and #5–#8 are all the same first five. That is deliberate — it is what makes legs comparable across configurations and clusters — but it has a sharp edge: legs sharing prompts also share the LLM gateway's completion cache, so they must be spaced (§7.4).

> `max_tasks` **above the benchmark's task pool is not an error.** The runner asks the MCP for the task list and slices it — `task_ids[:max_tasks]` — so a request for more tasks than exist yields the whole pool with no warning, and `summary.total` then reports the pool size rather than what you asked for. There is no upper bound on the field to catch it. The pools differ by an order of magnitude (gsm8k **1,319** — the HuggingFace `test` split, not the 8.5K dataset — tau2 **114** in the default `retail` domain, appworld **168** in the `test_normal` split), so check the size before requesting a large run: `BENCHMARKS_PRIMER.md` has them in one table. Selection is **deterministic** — the first `max_tasks` of the pool — which is what makes a smaller run's tasks a prefix of a larger one's.

> **Timeout tuning (learned e2e):** tau2 with `max_tasks=20 max_parallel_sessions=4` exceeded the 900s default and failed; re-running with `timeout_seconds=1800` succeeded (~915s).

6.4 Poll run status / list runs

```
# One run's full state (status, summary, per-task results, artifacts once exported).
curl -s "$SVC/benchmarks/gsm8k/runs/$RUN" -H "Authorization: Bearer $TOKEN" | python3 -m
   json.tool
# All runs for this benchmark scoped to your instance.
curl -s "$SVC/benchmarks/gsm8k/runs" -H "Authorization: Bearer $TOKEN" | python3 -m json.tool
```

`status` moves `pending → running → succeeded|failed`. On success, `summary` carries `{total, succeeded, evaluated_pass, pass_rate, wall_seconds}`. A simple poll loop:

```
# Terminal statuses are succeeded | failed | error | cancelled — a loop that waits only for
# succeeded/failed spins forever on the other two.
while :; do
  S=$(curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/runs/$RUN" -H "Authorization: Bearer $TOKEN" \
      | python3 -c 'import sys,json; print(json.load(sys.stdin)["status"])')
  echo "status=$S"
  case "$S" in succeeded|failed|error|cancelled) break ;; esac
  sleep 10
done
# The one-line verdict.
curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/runs/$RUN" -H "Authorization: Bearer $TOKEN" >
   /tmp/run.json
python3 - <<'EOF'
import json
d = json.load(open("/tmp/run.json")); s = d["summary"]
print("%s  pass_rate=%s  %s/%s  wall=%.0fs" % (
    d["status"], s["pass_rate"], s["evaluated_pass"], s["total"], s["wall_seconds"]))
EOF
```

6.5 Get results (report)

Two report views, both reading structured records from MLflow (return `409` if MLflow isn't configured for the instance — see §4.5):

```
# Per-run report: records filtered to this run's session ids, plus its S3 artifacts.
curl -s "$SVC/benchmarks/gsm8k/runs/$RUN/report" -H "Authorization: Bearer $TOKEN" | python3 -m
   json.tool
# Experiment rollup: time-windowed aggregates across an experiment's runs.
curl -s "$SVC/benchmarks/gsm8k/report?experiment=default&window_h=3" \
  -H "Authorization: Bearer $TOKEN" | python3 -m json.tool
```

Per-run report is `{run_id, benchmark, experiment, trace_count, records[], artifacts[]}`; each record (`MLflowTraceRecord`) has per-session timing breakdown, LLM/tool latencies + token counts, infra CPU/mem, and the evaluation outcome.

6.6 Download output files from S3

When the instance has an S3 `bucket` set, each completed run is exported (fail-soft) and its objects appear in the run state under `artifacts[]` (and in the per-run report). Layout:

```
<prefix>/<preferred_username>/<encoded-iss>/<benchmark>/<run_id>/
    run.json                # the run summary (always written)
    report.ndjson           # one record per task (empty if MLflow unconfigured)
    report.parquet          # analytics-friendly (only when there are records)
    token_report.ndjson     # lean per-task token view, projected from the records
    token_report.parquet
    span_report.ndjson      # one row per OTEL span per task, incl. the span name
    span_report.parquet
    manifest.json           # self-describing index of the above (never lists itself)
```

`span_report.*` is the evidence layer under the other two: `report` / `token_report` give per-task *counts*, `span_report` gives the spans those counts were derived from, so "did this task really make one model call,

or did we lose a span?" is answerable from the artifacts alone. Its columns are a fixed whitelist (`mlflow_report.SPAN_ROW_KEYS`) — span attributes can carry prompts, and these objects are public-read, so nothing outside that list is published.

Which process named a span. The tree is stitched from two processes by W3C trace propagation (`runner/a2a_agent.py` injects the context into the A2A request headers), so `depth` is depth in the *merged* trace, not a process boundary. The split is clean anyway: the **Service** names exactly four spans — `Agent.Session` (`kind=root`) and `MCP.CreateSession` / `Agent.Call` / `Evaluator.Evaluate` (`kind=phase`) — and emits nothing else, because it carries no OTEL auto-instrumentation. Everything at **depth ≥ 2 comes from inside the agent pod**: the spans its runtime names (`invoke_agent`, `chat <model>`, `execute_tool <tool>`) plus the A2A/ASGI internals that dominate the count. On one gsm8k task, 4 of 96 spans were the Service's and the other 92 came from the agent pod. That gives a free measurement worth knowing: the agent-side `POST /` span brackets the whole agent-side task, so `Agent.Call – POST /` is the Service's own per-task cost — a median **16.3 ms of a 2.5 s gsm8k task, 15.9 ms of a 40 s tau2 task**.

Error text is classified, not published

`span_report.*`'s whitelist covers the artifact whose fields come from third-party spans. Two fields in the *other* artifacts were free-text pass-throughs in the same way, and are handled by the companion discipline:

artifact	field	source
`run.json`	`error`, and every `results[].error`	`RunState`
`report.ndjson` / `report.parquet` (and, projected, `token_report.*`)	`status_message`	`MLflowTraceRecord`

Whatever the agent, the MCP server or the LLM gateway put in a message landed in the bucket unfiltered, and across the runs published before this change that included internal hostnames (the LLM gateway's among them), a team UUID and a spend figure. Every image that carries `src/autobench/public_errors.py` instead writes

```
<category> (shape <8 hex>)        e.g.  mcp_connect_timeout (shape 4f1c92ab)
```

- **category** — a member of the closed set in `src/autobench/public_errors.py` (`CATEGORIES`). Nothing outside that set is ever written, so an upstream message nobody anticipated is at worst `other`; when that happens the Service logs the verbatim text at WARNING so a bucket can be added.
- **shape** — `sha256` truncated to 8 hex over the message *template*: URLs, UUIDs, hostnames, emails, keys and digits are replaced with placeholders **before** hashing. Tasks that failed the same way therefore share a shape id and group, while a guessed hostname cannot be confirmed by reconstructing the string and re-hashing it.

This is classification, not redaction, on purpose — a redactor has to enumerate every bad pattern, and only has to miss one. The substitution happens in `s3_export.export_run`, the single point every artifact passes through; the in-memory models keep the original, so **the authenticated API (`/runs/{id}`, `/report`) and the pod log still return the verbatim message**. Only the S3 projection is classified. Runs published by an earlier image still hold raw strings, so anything reading both eras should go through `reference/causelib.py`, which parses a classified value and falls back to the legacy patterns.

The S3 URL is fully determined — you can construct it

Objects are addressed as `<url-root>/<key>`, and the key is the layout above. For the bucket these runs use:

URL root	`https://rossoctl-benchmarking.s3.us-east-1.amazonaws.com/`
Key	`<s3.prefix>/<preferred_username>/<encoded-iss-host>/<benchmark>/<run_id>/<artifact>`

So a real, working URL — anonymously readable, no credentials, no AWS CLI:

```
https://rossoctl-benchmarking.s3.us-east-1.amazonaws.com/ykt3/benchmarker/keycloak-
   keycloak.apps.ykt3.hcp.res.ibm.com-realms-rossoctl/gsm8k/20261012195705-
   8bd41896/report.ndjson
```

`s3.prefix` is per instance (`ykt3/` on the OpenShift instance, `kind/` on KinD), and the issuer host is scheme-stripped with `.` / `:` / `/` replaced by `-`. **The bucket is public and anonymously listable**, so treat everything exported as published.

```
# The authoritative list — name, format, size and the full URL, straight from the run state.
curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/runs/$RUN" -H "Authorization: Bearer $TOKEN" >
   /tmp/run.json
python3 - <<'EOF'
import json
for a in json.load(open("/tmp/run.json"))["artifacts"]:
    print("%-8s %9d  %s" % (a["format"], a["size_bytes"], a["url"]))
EOF
# Download every artifact by URL — plain curl, no AWS credentials needed.
PREFIX=$(curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/runs/$RUN" -H "Authorization: Bearer $TOKEN"
   \
  | python3 -c 'import sys,json; print(json.load(sys.stdin)["artifacts_prefix"])')
mkdir -p "/tmp/autobench/$PREFIX" && cd "/tmp/autobench/$PREFIX"
for f in run.json report.ndjson report.parquet token_report.ndjson token_report.parquet \
         span_report.ndjson span_report.parquet manifest.json; do
  curl -sS -O "https://rossoctl-benchmarking.s3.us-east-1.amazonaws.com/$PREFIX/$f"
done && ls -la
# Or discover the set from the manifest, which indexes every object but itself.
curl -s "https://rossoctl-benchmarking.s3.us-east-1.amazonaws.com/$PREFIX/manifest.json" \
  | python3 -m json.tool
# Private bucket instead of public_read? Then use the AWS CLI with credentials.
aws s3 cp "s3://rossoctl-benchmarking/$PREFIX/" ./ --recursive
```

6.7 Tear down

```
curl -s -X DELETE "$SVC/benchmarks/gsm8k/deploy?
   namespace=team1&agent=tool_calling&experiment=default" \
  -H "Authorization: Bearer $TOKEN" -o /dev/null -w '%{http_code}\n'   # -> 204
```

Deletes the agent and the shared tool. For a model-swap variant deployed via `POST /agents`, delete it with `DELETE /agents/{namespace}/{name}` (the shared tool is left for other experiments).

6.8 Comparing gateway × model combinations

Model, gateway and key are all **deploy-time** settings, so one Service can measure several combinations without a reinstall. Each run deploys one combination, and its artifacts say which.

1. **Put each extra gateway's key in its own Secret**, once, in the workload namespace (Admin Guide §3.3). The Service has no Kubernetes access and never handles the key.
2. **Run each combination as its own deploy and run.** Give each gateway a label, which is what the artifacts carry:

   ```
   # the instance's own gateway and default model
   autobench-cli all --benchmark gsm8k --tasks 5 --llm-gateway ete-ext --teardown --mirror
      /tmp/autobench
   # another gateway, its key Secret, and a model that key is granted
   autobench-cli all --benchmark gsm8k --tasks 5 --teardown --mirror /tmp/autobench \
     --llm-api-base https://litellm.cb.example.com --llm-key-secret cb-key --llm-gateway cb \
     --model openai/rits/zai-org/glm-5-3
   ```

3. **Compare:** `python3 reference/gen-gateway-compare.py /tmp/autobench -o /tmp/compare.md`. It writes one row per (benchmark, gateway, model):
   - pass rate;
   - tokens per task, and tasks lost before their first model call;
   - $/task and $/pass at that gateway's posted rates;
   - median latency.

What the artifacts record:

- **`run.json`** carries `llm: {gateway, model, key_secret}`.
- **Every `report.ndjson` / `token_report.*` row** carries `llm_gateway`. `model` is the model the agent's own spans report it actually called.
- **The gateway URL appears nowhere.** The bucket is public.

What to know:

- **Check the gateway from the cluster, not from your laptop.** A gateway's front end may allowlist source addresses. KinD egresses through your machine; an OpenShift cluster does not.
  - An **HTML `403 Forbidden`** from `GET <base>/v1/models`, with or without the key, means the front end refused the cluster's address, and every task fails in well under a second.
  - A **JSON `401`** without the key means the request reached LiteLLM.
  - Probe from a pod in the cluster, e.g. `kubectl exec` into the Service pod and fetch the URL with Python.
- **The model must be one that key is granted.** `GET /v1/models` with that key lists its grants, not the gateway's whole catalogue.
- **Prices are per gateway.** `reference/model_prices.json` → `gateways.<label>.models`.
  - A labelled gateway with its own card is priced only from that card, and anything else on it shows **unpriced**, never another gateway's price.
  - A label with no card uses the top-level card, which is the internal ETE gateway's.

- **One combination at a time per benchmark and namespace.** The MCP tool's name has no experiment suffix, so two deployments of one benchmark share it, and with it the user simulator's gateway on tau2. `all` tears down before it deploys.
- **What follows the agent onto the gateway:** tau2's user simulator does. The IBAC judge does **not**, because it is cluster-wide, so plugin legs still pay the cluster's judge gateway.
- **The run learns the gateway from the Service's memory of the deploy.** If the Service restarted between deploy and run, `llm` and `llm_gateway` are `null`; they are never guessed.
- **The CLI trusts the deploy echo, not the status code.** If `--llm-*` comes back without an `llm` echo, it stops with exit code `3`: the Service predates v1.36 and ignored the override. It also refuses `--llm-*` / `--model` with `--no-deploy`, where they would have no effect.
- **Gateway caches:** each gateway has its own completion cache, so `BM_CACHE_GAP` (§7.4) spaces legs per gateway.

Measured: the whole 12-run on cb and on ETE-Ext (KinD)

The canonical matrix (§7.4), run twice on one KinD node on 2026-10-12, changing only the gateway and its models:

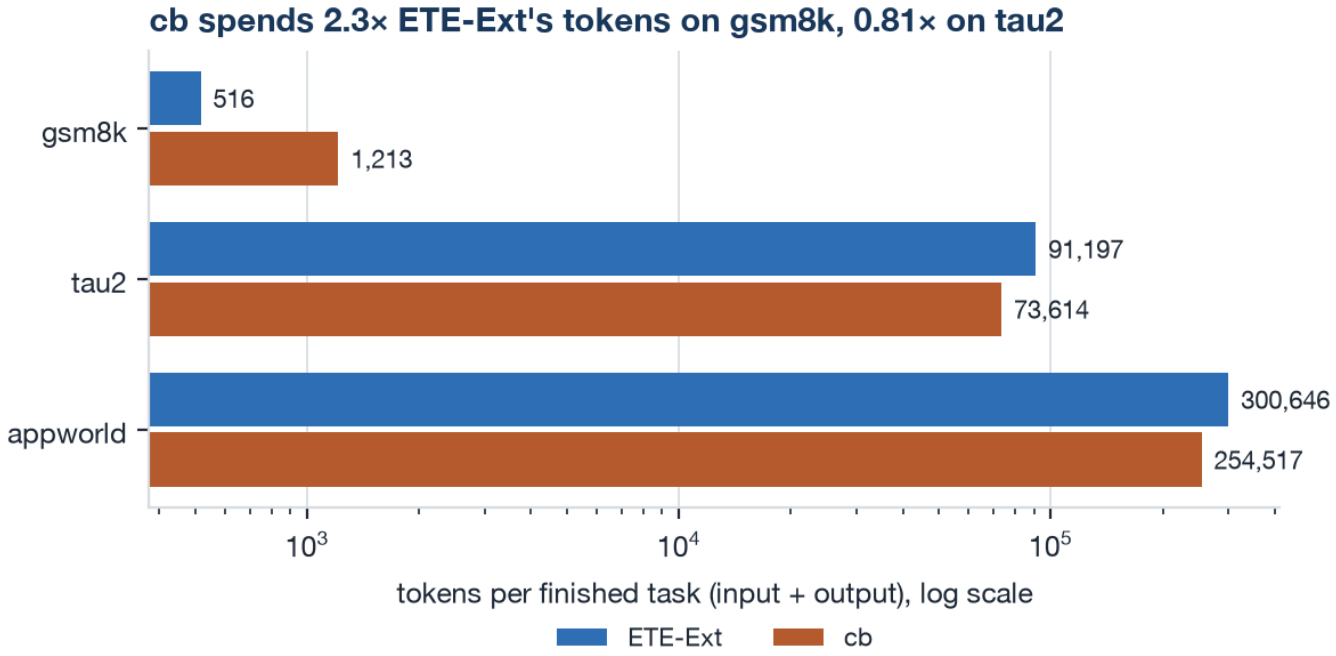
- **ETE-Ext:** the published v1.36 baseline's own models.
- **cb:** a second LiteLLM gateway serving self-hosted models, with `rits/zai-org/glm-5-3` on every leg and `NVIDIA-Nemotron-3-Ultra` on #4, the model-swap leg.

Both runs kept the IBAC judge on ETE-Ext. Each deploy named its gateway by label and its key by Secret name (`cb-key`), so neither the URL nor the key appears in any artifact or here.

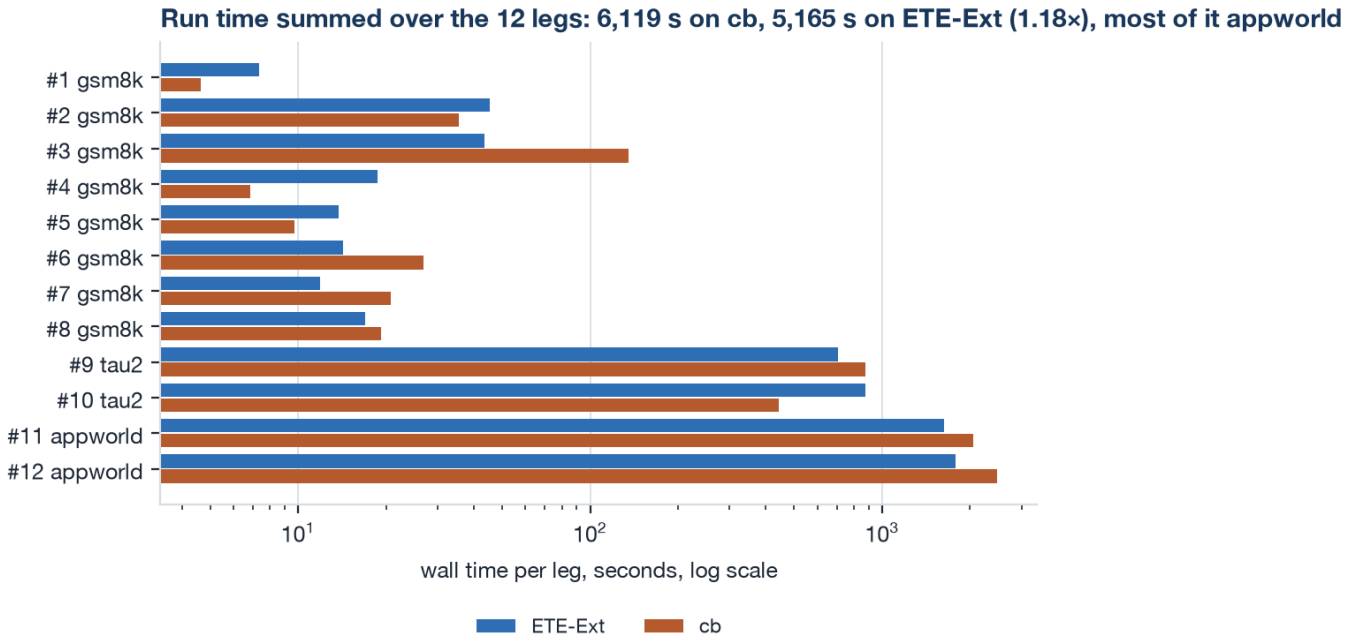
leg	benchmark	model on ETE-Ext	model on cb	pass ETE-Ext	pass cb	tasks finished ETE-Ext	tasks finished cb
#1	gsm8k	`Azure/gpt-5-mini-2025-08-07`	`rits/zai-org/glm-5-3`	1.00	1.00	1 / 1	1 / 1
#2	gsm8k	`Azure/gpt-5-mini-2025-08-07`	`rits/zai-org/glm-5-3`	1.00	1.00	10 / 10	10 / 10
#3	gsm8k	`Azure/gpt-5-mini-2025-08-07`	`rits/zai-org/glm-5-3`	1.00	0.98	50 / 50	50 / 50
#4	gsm8k	`Azure/gpt-4.1`	`rits/nvidia/NVIDIA-Nemotron-3-Ultra-550B-A55B-NVFP4`	0.80	0.80	5 / 5	5 / 5
#5	gsm8k	`Azure/gpt-5-mini-2025-08-07`	`rits/zai-org/glm-5-3`	1.00	1.00	5 / 5	5 / 5
#6	gsm8k	`Azure/gpt-5-mini-2025-08-07`	`rits/zai-org/glm-5-3`	1.00	1.00	5 / 5	5 / 5

leg	benchmark	model on ETE-Ext	model on cb	pass ETE-Ext	pass cb	tasks finished ETE-Ext	tasks finished cb
#7	gsm8k	`Azure/gpt-5-mini-2025-08-07`	`rits/zai-org/glm-5-3`	1.00	1.00	5 / 5	5 / 5
#8	gsm8k	`Azure/gpt-5-mini-2025-08-07`	`rits/zai-org/glm-5-3`	1.00	1.00	5 / 5	5 / 5
#9	tau2	`aws/claude-sonnet-5`	`rits/zai-org/glm-5-3`	0.90	1.00	10 / 10	10 / 10
#10	tau2	`aws/claude-sonnet-5`	`rits/zai-org/glm-5-3`	0.80	0.85	20 / 20	20 / 20
#11	appworld	`gemini-2.5-pro`	`rits/zai-org/glm-5-3`	0.00	0.00	4 / 5	3 / 5
#12	appworld	`gemini-2.5-pro`	`rits/zai-org/glm-5-3`	0.00	0.00	19 / 20	9 / 20

leg	input ETE-Ext	input cb	output ETE-Ext	output cb	total ETE-Ext	total cb	wall ETE-Ext	wall cb	cost ETE-Ext	cost cb
#1 gsm8k	320	826	150	184	470	1,010	7 s	5 s	$0.00038	$0.00
#2 gsm8k	3,166	9,276	1,629	1,587	4,795	10.9 K	45 s	36 s	$0.00405	$0.00
#3 gsm8k	15.7 K	44.8 K	9,166	21.5 K	24.9 K	66.3 K	44 s	135 s	$0.0223	$0.00
#4 gsm8k	3,536	4,303	292	225	3,828	4,528	19 s	7 s	$0.00941	$0.00
#5 gsm8k	1,564	4,550	1,071	1,006	2,635	5,556	14 s	10 s	$0.00253	$0.00
#6 gsm8k	1,564	4,565	1,007	752	2,571	5,317	14 s	27 s	$0.00241	$0.00
#7 gsm8k	1,564	4,526	815	942	2,379	5,468	12 s	21 s	$0.00202	$0.00
#8 gsm8k	1,564	4,520	1,327	804	2,891	5,324	17 s	19 s	$0.00304	$0.00
#9 tau2	965.1 K	815.9 K	22.9 K	28.6 K	988.0 K	844.5 K	705 s	879 s	$1.64	$0.00
#10 tau2	1.71 M	1.31 M	38.1 K	52.4 K	1.75 M	1.36 M	877 s	441 s	$2.89	$0.00
#11 appworld	911.1 K	682.0 K	96.7 K	77.5 K	1.01 M	759.5 K	1,629 s	2,058 s	$2.11	$0.00
#12 appworld	5.38 M	2.05 M	529.4 K	249.0 K	5.91 M	2.29 M	1,782 s	2,482 s	$12.02	$0.00
all 12 legs	**8.99 M**	**4.93 M**	**702.6 K**	**434.5 K**	**9.70 M**	**5.37 M**	**5,165 s**	**6,119 s**	**$18.70**	**$0.00**



cb spends 2.3× ETE-Ext's tokens on gsm8k, 0.81× on tau2.



Run time summed over the 12 legs: 6,119 s on cb, 5,165 s on ETE-Ext (1.18×), most of it appworld.

How to read it:

- **Tokens and wall time cover the tasks that finished.** A task killed by the 600 s per-task budget leaves no `report.ndjson` row, so its tokens are missing here and its leg's pass rate counts it as failed. appworld lost 2 of 25 tasks that way on ETE-Ext and 13 of 25 on cb. That is most of why cb's appworld totals are lower: per finished task the two are much closer (254,517 tokens on cb against 300,646 on ETE-Ext).
- **The model changes with the gateway on tau2 and appworld**, so those rows compare gateway × model combinations, not gateways alone. Only #4 swaps on purpose.

- **Dollars are posted rates** from `reference/model_prices.json`, per gateway. cb's models are self-hosted (`hosted_vllm`) with no per-token price, hence $0.00; its own compute is not in that figure.
- **The IBAC judge stayed on ETE-Ext's `Azure/gpt-4.1`** for legs #6–#8 on both runs; its calls are billed but not in the agent's telemetry (§2), so neither cost column includes them.
- **Wall time is that day's.** Both runs were on one KinD node on 2026-10-12 (local time; the run ids are UTC), one after the other, so the comparison is fair between them; across days the gateways' latency moves (§2).

leg	run id on ETE-Ext	run id on cb
#1	`20261012155448-8bb425fa`	`20261013003659-dfe8ddd5`
#2	`20261012162903-83d5f1e9`	`20261013012241-7655a2ae`
#3	`20261012164913-3a195d6e`	`20261013013959-67135ee4`
#4	`20261012174041-489ca9c7`	`20261013024245-ba983bcc`
#5	`20261012172122-e5a5e97e`	`20261013022045-487d12ed`
#6	`20261012173846-01c899df`	`20261013024032-e3ea02da`
#7	`20261012175611-e291172d`	`20261013025814-a1db3383`
#8	`20261012193337-2817edb7`	`20261013031549-76996256`
#9	`20261012172401-e3495925`	`20261013022336-db7a8bd7`
#10	`20261012163248-11a97e56`	`20261013012546-c492e2c1`
#11	`20261012165157-39de49ae`	`20261013014413-788ab836`
#12	`20261012155722-74b7895f`	`20261013003858-bb546fd9`

7. End-to-end examples (validated flows)

What you need on the client side

Python version. The client side is deliberately undemanding: §7.0 needs only `curl` plus any `python3` for JSON formatting, and `autobench-cli` (§7.1) imports nothing beyond the standard library. Versions we have actually run:

Python	Where it was used	Result
3.12.12	recommended for a client venv — matches the workload images exactly	✅
3.12.14	the Service container itself (`FROM python:3.12-slim`)	✅
3.11.14	the repo's own venv; the declared floor (`requires-python = ">=3.11"`)	✅
3.14.3	the laptop that drove both 12-run matrices	✅

Anything **≥ 3.11** is fine. 3.12.12 is the tidiest choice because it is exactly what the exgentic agent and MCP images run, so your driver matches the workload interpreter — but nothing in the client depends on it, and the `curl` path in §7.0 is version-insensitive.

Do you have to clone the repo? For **§7.0, no** — it is `curl` and `python3` only, so a token and a Service URL are all you need. For **§7.1** you need the package that provides the `autobench-cli` entry point, but a manual

clone is still optional: the repository is public, so install it straight from git.

Installing it as a tool is the shortest path, because you want a *command*, not a library:

```
# No clone, no venv to activate. Puts `autobench-cli` (and `autobench-service`) on your PATH.
uv tool install "rossoctl-autobench @ git+https://github.com/rossoctl/autobench"
autobench-cli --help
# Or run it without installing anything at all:
uvx --from "rossoctl-autobench @ git+https://github.com/rossoctl/autobench" autobench-cli --help
```

`uv tool install` builds an isolated environment under `$(uv tool dir)` and links only the console scripts into `~/.local/bin`, so the Service's dependencies never land anywhere you import from. Two consequences worth knowing:

- **`~/.local/bin` has to be on your PATH.** `uv tool update-shell` adds it; `uv tool list` shows what is installed and which executables it provides.
- **There is no `--no-deps` here, and you do not want one.** `uv tool install` rejects the flag outright, and skipping fastapi/boto3/pydantic would only leave the `autobench-service` script broken — the isolated environment is already the thing `--no-deps` was protecting you from.

uv caches the resolved git revision, so a second `uv tool install` of the same URL can be a no-op even after upstream moves. Force it:

```
uv tool install --force --reinstall --refresh \
  "rossoctl-autobench @ git+https://github.com/rossoctl/autobench"
uv tool list                    # the version it resolved to (the CLI has no --version flag)
```

Pin instead of chasing `main` when you want reproducibility — append `@<tag-or-sha>` to the URL (`...autobench@v1.36`: each release tag marks the commit that pins its Service image).

Or into a venv, if you are also developing against it

```
uv venv --python 3.12.12 && source .venv/bin/activate
# Point at the repo, NOT at `.` — `-e .` looks for pyproject.toml in the CURRENT directory, so
# running it from a scratch/testbed folder fails with "does not appear to be a Python project".
uv pip install -e /path/to/autobench --no-deps        # -e: git pull updates the CLI in place
# No clone, into the active venv — same as the tool install but scoped to this venv:
uv pip install "rossoctl-autobench @ git+https://github.com/rossoctl/autobench" --no-deps
```

Here `--no-deps` *is* worth passing — the CLI is stdlib-only, and the alternative adds fastapi, boto3, pydantic and the MCP/A2A SDKs to a venv you may be using for other things. Drop it if you also want to run the Service or its test suite from that venv.

Two failure modes this path has actually produced:

- **`uv pip install` installs into an environment, not onto your PATH.** With no venv active it falls back to one it picks itself (`~/.venv` when you run it from `$HOME`), and nothing is linked into `~/.local/bin` — the install "succeeds" and `autobench-cli` is still `command not found`. `uv tool install` is the fix.
- **`Audited 1 package` means nothing happened.** That is uv reporting the requirement was already satisfied, not a fresh build; it is not confirmation that you picked up a new commit.

Keep any venv outside a cloud-synced folder (Box/Dropbox/iCloud) — a venv is thousands of small files, and sync tools are slow and occasionally destructive with it.

Where the source ends up

The `autobench-cli` on your PATH is **not** the source — it is a generated shim that does `from autobench.cli import main`. Where the real `cli.py` lives depends on how you installed, which also decides whether a `git pull` reaches you:

Install	shim in	`cli.py` lives in	Picks up upstream changes?
`uv tool install "… @ git+https://…"`	`~/.local/bin/`	`$(uv tool dir)/rossoctl-autobench/lib/python3.X/site-packages/autobench/` (a **copy**, built from one commit)	No — pinned, and the git revision is cached. `--force --reinstall --refresh`
`uv tool install --editable /path/to/autobench`	`~/.local/bin/`	`/path/to/autobench/src/autobench/` (the clone)	Yes, immediately
`uv pip install "…@ git+https://…"`	`<venv>/bin/` — **on your PATH only while that venv is active**	`<venv>/lib/python3.X/site-packages/autobench/` (a copy)	No — pinned. Re-run with `--reinstall`
`uv pip install -e /path/to/autobench`	`<venv>/bin/`, same caveat	`/path/to/autobench/src/autobench/` (the clone; only a finder hook is installed)	Yes, immediately
no install, `PYTHONPATH=<repo>/src python -m autobench.cli`	—	the clone itself	Yes, immediately

Never guess the path — ask the interpreter that is actually running it. For a venv install, that is the venv's own `python`; for a tool install the package is deliberately *not* importable from your shell's python, so ask uv instead:

```
python -c "import autobench.cli as m; print(m.__file__)"     # venv installs
uv tool list                                                  # tool installs: version +
   executables
head -2 "$(command -v autobench-cli)"                         # the shim names its interpreter
```

Note that every option installs the **whole distribution**, server modules included (`app.py`, `routes/`, `runner/`, `s3_export.py`), because they ship together. On the `uv pip install --no-deps` path the client half still works while `import autobench.app` does not — the third-party dependencies were skipped, not the files. That the client keeps working is the point, and you can confirm it:

```
python -c "
import autobench.cli, sys
print([m for m in ('fastapi','httpx','boto3','pydantic') if m in sys.modules] or 'no server deps
   loaded')"
```

7.0 Run #1 start to finish, on ykt3 (OpenShift)

Run #1 of the canonical matrix: gsm8k, 1 task, no gateway, no plugins. The `run_id` and numbers are from leg #1 of the published v1.36 OpenShift matrix (`docs/results/v1.36-2026-10-12/`), which made these same API calls through `reference/run-12.py` . Paste the block, then the steps.

```
# ---- 0. environment (the Service and its workloads both on ykt3, team1) ----
export SVC="https://autobench-rossoctl-system.apps.ykt3.example.com"
export KC="https://keycloak-keycloak.apps.ykt3.example.com"   # the instance's iss origin
export REALM=rossoctl KC_CLIENT=rossoctl KC_USER=benchmarker
export BENCH=gsm8k
export SCOPE="namespace=team1&agent=tool_calling&experiment=default"
export CURL_OPTS=-k          # ykt3's route serves a self-signed cert
read -rs -p "benchmarker password: " KC_PASS; echo      # never put this in a file you commit
# ---- 1. caller token ----
export TOKEN=$(curl -s $CURL_OPTS "$KC/realms/$REALM/protocol/openid-connect/token" \
  -d grant_type=password -d client_id="$KC_CLIENT" \
  -d username="$KC_USER" -d password="$KC_PASS" \
  | python3 -c 'import sys,json; print(json.load(sys.stdin)["access_token"])')
unset KC_PASS
curl -s $CURL_OPTS "$SVC/hello" -H "Authorization: Bearer $TOKEN" | python3 -m json.tool
#   -> {"iss": ".../realms/rossoctl", "preferred_username": "benchmarker", ...}
# ---- 2. deploy the MCP tool + A2A agent (idempotent: delete first) ----
curl -s $CURL_OPTS -X DELETE "$SVC/benchmarks/$BENCH/deploy?$SCOPE" \
  -H "Authorization: Bearer $TOKEN" -o /dev/null -w 'delete -> %{http_code}\n'   # 204 or 404
sleep 10
curl -s $CURL_OPTS -X POST "$SVC/benchmarks/$BENCH/deploy" \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d '{"agent":"tool_calling","namespace":"team1","experiment":"default"}' \
  -o /dev/null -w 'deploy -> %{http_code}\n'                                    # 201
# ---- 3. wait until BOTH are Ready, four polls in a row ----
# One Ready reading is not enough: an agent whose MCP is not serving yet exits and
# CrashLoopBackOffs, so readiness flaps. A run accepted mid-flap records 0 tasks.
STABLE=0; until [ "$STABLE" -ge 4 ]; do
  R=$(curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/status?$SCOPE" -H "Authorization: Bearer
   $TOKEN" \
      | python3 -c 'import sys,json; d=json.load(sys.stdin); print(int(bool(d["tool_ready"] and
   d["agent_ready"])))')
  [ "$R" = 1 ] && STABLE=$((STABLE+1)) || STABLE=0
  echo "ready=$R stable=$STABLE"; sleep 10
done
sleep 15     # settle: the agent reports Ready a moment before it is serving (run-12.py does the
   same)
# ---- 4. submit the run ----
export RUN=$(curl -s $CURL_OPTS -X POST "$SVC/benchmarks/$BENCH/runs" \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d '{"agent":"tool_calling","namespace":"team1","experiment":"default",
       "max_tasks":1,"max_parallel_sessions":1,"timeout_seconds":300}' \
  | python3 -c 'import sys,json; print(json.load(sys.stdin)["run_id"])')
echo "run_id=$RUN"                                  # e.g. 20261012195705-8bd41896
# ---- 5. poll to a terminal status, then print the verdict ----
while :; do
  S=$(curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/runs/$RUN" -H "Authorization: Bearer $TOKEN" \
      | python3 -c 'import sys,json; print(json.load(sys.stdin)["status"])')
  echo "status=$S"; case "$S" in succeeded|failed|error|cancelled) break ;; esac; sleep 10
done
#   -> succeeded, summary
   {"total":1,"succeeded":1,"evaluated_pass":1,"pass_rate":1.0,"wall_seconds":6.7}
#      Note 96 spans for ONE gsm8k task: 88 are A2A/HTTP framework internals.
```

```
# ---- 6. artifacts: list the S3 URLs, then mirror them locally ----
export PREFIX=$(curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/runs/$RUN" -H "Authorization: Bearer
    $TOKEN" \
  | python3 -c 'import sys,json; print(json.load(sys.stdin)["artifacts_prefix"])')
echo "$PREFIX"
#   -> ykt3/benchmarker/keycloak-keycloak.apps.ykt3.hcp.res.ibm.com-realms-
    rossoctl/gsm8k/20261012195705-8bd41896
mkdir -p "/tmp/autobench/$PREFIX" && (cd "/tmp/autobench/$PREFIX" && \
  for f in run.json report.ndjson report.parquet token_report.ndjson token_report.parquet \
           span_report.ndjson span_report.parquet manifest.json; do
    curl -sS -O "https://rossoctl-benchmarking.s3.us-east-1.amazonaws.com/$PREFIX/$f"; done &&
    ls -la)
# ---- 7. analyse: per-task tokens, and whether the telemetry is trustworthy ----
cd "/tmp/autobench/$PREFIX"
python3 - <<'EOF'
import json
rows = [json.loads(l) for l in open("report.ndjson") if l.strip()]
print("%-6s %-38s %7s %7s %5s %6s  %s" % ("task", "model", "in", "out", "llm", "tool",
    "passed"))
for r in sorted(rows, key=lambda x: str(x["task_id"])):
    print("%-6s %-38s %7d %7d %5s %6s  %s" % (
        r["task_id"], r["model"], r["llm_input_tokens"], r["llm_output_tokens"],
        r["llm_count"], r["tool_count"], r["evaluation_result"]))
# llm<=1 alongside tool>=2 is impossible -- every tool call needs a preceding model turn -- so
    it
# means the usage-bearing span was lost and the token counts understate the real cost.
bad = [r for r in rows if (r["llm_count"] or 0) <= 1 and (r["tool_count"] or 0) >= 2]
print("lost token attribution:", len(bad), "of", len(rows))
EOF
#   -> 0      openai/Azure/gpt-5-mini-2025-08-07         320      86     1      1  True
#      lost token attribution: 0 of 1
# Which spans did the task actually invoke? A healthy task shows one chat span per model call
# (llm_count above). Fewer chat spans than calls means a usage-bearing span was lost.
python3 - <<'EOF'
import collections, json
rows = [json.loads(l) for l in open("span_report.ndjson") if l.strip()]
print("spans:", len(rows))
for kind, n in collections.Counter(r["kind"] for r in rows).most_common():
    print("  %4d  %s" % (n, kind))
for r in rows:
    if r["kind"] == "chat":
        print("  chat  max_tokens=%s  in=%s  out=%s" % (
            r["request_max_tokens"], r["input_tokens"], r["output_tokens"]))
EOF
#   -> spans: 96  (other 88, phase 3, tool 2, root 1, agent 1, chat 1)
#      chat  max_tokens=None  in=320  out=86     <- the one model call
# ---- 8. tear down ----
curl -s $CURL_OPTS -X DELETE "$SVC/benchmarks/$BENCH/deploy?$SCOPE" \
  -H "Authorization: Bearer $TOKEN" -o /dev/null -w 'teardown -> %{http_code}\n'   # 204
```

7.1 The same thing in one command (`autobench-cli`)

`autobench-cli` is a stdlib-only client that performs §7.0 end to end — token, pre-clean, deploy, the readiness-stability gate, run, 424 retry, poll, artifact listing and local mirror — and exits non-zero if the run did not succeed.

```
export BM_BASE="https://autobench-rossoctl-system.apps.ykt3.example.com"
export BM_ISS="https://keycloak-keycloak.apps.ykt3.example.com/realms/rossoctl"
```

```
export BM_PASSWORD_FILE="$HOME/.rossoctl-ykt3/benchmarker.pass"     # chmod 600
export BM_INSECURE=1                                                # the Route's edge TLS cert
# See "What you need on the client side" above for install options; the shortest is:
#   uv tool install "rossoctl-autobench @ git+https://github.com/rossoctl/autobench"
autobench-cli all --benchmark gsm8k --tasks 1 --timeout 300 --mirror /tmp/autobench
```

Which prints, for a v1.35 run on ykt3 (`--insecure` here stands in for `BM_INSECURE=1`):

```
[15:20:29] run_id=20261004192029-5beb2676
[15:20:40] terminal=succeeded summary={"total": 1, ..., "pass_rate": 1.0, "wall_seconds": 3.1}
…          artifacts_prefix=ykt3/benchmarker/keycloak-...-rossoctl/gsm8k/20261004192029-5beb2676
  json           469  https://rossoctl-benchmarking.s3.us-east-1.amazonaws.com/.../run.json
  ...
…          mirrored 8/8 -> /tmp/autobench/ykt3/.../20261004192029-5beb2676
  -rw-r--r--      2882  Oct 04 15:20  manifest.json
  -rw-r--r--      1023  Oct 04 15:20  report.ndjson
  -rw-r--r--     11984  Oct 04 15:20  report.parquet
  -rw-r--r--       469  Oct 04 15:20  run.json
  -rw-r--r--     49972  Oct 04 15:20  span_report.ndjson
  -rw-r--r--     10394  Oct 04 15:20  span_report.parquet
  -rw-r--r--       370  Oct 04 15:20  token_report.ndjson
  -rw-r--r--      4425  Oct 04 15:20  token_report.parquet
  8 files, 81,519 bytes

  cd /tmp/autobench/ykt3/.../20261004192029-5beb2676

  status      succeeded
  pass_rate   1.0   (1/1)
  wall        3s
```

The trailing `cd` is there because the mirror path is absolute and deeply nested — retyping it relative to your current directory is the easy mistake. The sizes are also the quickest check that nothing arrived truncated. Feed that directory straight into §7.0 step 7 to analyse the run.

Each step is also a subcommand, so the CLI doubles as a way to see one HTTP call at a time:

```
autobench-cli whoami                                  # GET /hello
autobench-cli list                                    # GET /benchmarks
autobench-cli mlflow-health                           # GET /mlflow/health (§5.1); exit 8 = not
   healthy
autobench-cli deploy    --benchmark gsm8k
autobench-cli wait      --benchmark gsm8k             # the readiness-stability gate
autobench-cli run       --benchmark gsm8k --tasks 1    # prints the run_id
autobench-cli poll      --benchmark gsm8k --run "$RUN"
autobench-cli report    --benchmark gsm8k --run "$RUN" # GET …/report (needs MLflow)
autobench-cli artifacts --benchmark gsm8k --run "$RUN" --mirror /tmp/autobench  # + ls -l
autobench-cli teardown  --benchmark gsm8k
```

No install? It runs straight from a checkout too — `python -m autobench.cli all …` (the module imports nothing beyond the standard library, so the server's dependencies are not needed to drive the API).

Every option is catalogued in §7.2. Switching to KinD needs only three lines:

```
export BM_BASE="http://autobench.localtest.me:8080"
export BM_ISS="http://keycloak.localtest.me:8080/realms/rossoctl"
export BM_PASSWORD_FILE="$HOME/.rossoctl-kind/benchmarker.pass"; unset BM_INSECURE
```

> For the full 12-run matrix use `reference/run-12.py` instead — see §7.4. It implements the same gates independently; consolidating both behind `autobench.cli` is a known follow-up. If you want the 12 legs as 12 individual commands in the shape of the line above, they are tabulated in §7.5.

7.2 `autobench-cli` options

`autobench-cli <command> [options]`. The **command is positional** and required; everything else is a flag. Defaults below are the real argparse defaults, so a bare `autobench-cli all` runs one gsm8k task in `team1` with a 300s budget.

Commands. `all` runs the whole lifecycle; the rest are the individual steps, so the same tool serves "just run it" and "show me one HTTP call".

Command	Does
`all`	pre-clean → deploy → wait → run → poll → artifacts → summary (→ teardown if asked)
`whoami`	`GET /hello` — proves the token and shows which instance you routed to
`list`	`GET /benchmarks`
`mlflow-health`	`GET /mlflow/health` (§5.1) — run this **before** trusting a report. Exits 8 when any stage failed, so a bring-up script can gate on `$?`; `--no-round-trip` reads only
`deploy` / `teardown`	create / delete the MCP tool + A2A agent
`delete-agent` / `delete-tool`	`DELETE /agents/{ns}/{name}` / `/tools/{ns}/{name}` — one AgentRuntime by its exact `--name`; `404` counts as gone. `teardown` addresses a benchmark and experiment, and stops at a missing agent without reaching its MCP tool; these clean up what that leaves behind
`wait`	block on the readiness-stability gate
`run`	`POST …/runs`, prints the `run_id`
`poll`	follow one run to a terminal status
`report`	`GET …/report` (needs MLflow configured, else empty)
`artifacts`	list the run's S3 objects, and mirror them with `--mirror`

What to run, and how much of it.

Option	Default	Meaning
`--benchmark`	`gsm8k`	`gsm8k` | `tau2` | `appworld`
`--tasks N`	`1`	`max_tasks` — **how many** benchmark problems to attempt (capped by the pool, §6.3)
`--parallel N`	`1`	`max_parallel_sessions` — **how many at once**
`--timeout S`	`300`	`timeout_seconds` — wall budget for the **whole run**
`--task-timeout S`	*unset*	`task_timeout_seconds` — ceiling for **one** task, clamped to `--timeout`

`--tasks` and `--parallel` are independent: `--tasks 50 --parallel 4` is fifty problems, four concurrently. Task selection is **deterministic**, so `--tasks 1` runs the *same* problem every time — which is why it is the standard smoke test (a 1-task gsm8k run reproduces 320 input / 87 output tokens). Set `--task-timeout` on multi-turn work: without it one wedged task can consume the entire `--timeout`, which is why the canonical matrix gives tau2 600s per task under a 2100s wall. Keep even the smoke test at 300s: the gateway

occasionally stalls a single call, the agent cuts it at 120s (`REQUEST_TIMEOUT`, injected by the Service from image v1.35) and retries, and a 120s run budget leaves no room for that retry — a 1-task run that normally takes 3–10s then fails outright. §6.3 has the semantics in full — what a multi-task run is, and why both over- and under-asking on `--tasks` are silent.

Where it goes. `--namespace` (`team1`), `--agent` (`tool_calling`), `--experiment` (`default`) form the scope that `deploy` / `wait` / `teardown` address.

Deploy-time only — these bake into the workload, so they need a (re)deploy to take effect; a run cannot re-point them.

Option	Meaning
`--model`	override the agent's LLM, e.g. `openai/Azure/gpt-4.1`
`--preset`	AuthBridge `plugin_preset`: `auth-only` | `ibac-only` | `full`
`--plugin NAME:POLICY`	repeatable per-plugin override, e.g. `--plugin ibac:observe`
`--on-error`	chain-default policy: `enforce` | `observe` | `off`
`--llm-api-base URL`	another LLM gateway for this deployment; needs `--llm-gateway` (§6.8)
`--llm-key-secret NAME[:KEY]`	the Secret in `--namespace` holding that gateway's key (default `openai-secret:apikey`)
`--llm-gateway LABEL`	the label the artifacts publish for the gateway, never the URL

`all` modifiers.

Option	Effect
`--no-deploy`	skip the pre-clean **and** the deploy; reuse what is already there
`--teardown`	delete the deployment when finished

Note the asymmetry: `all` **pre-cleans by default** (an unconditional `DELETE` before deploying, so it is repeatable) but **does not tear down** — the deployment is left warm unless you ask.

Readiness gates. These exist because Service-reported readiness is not sufficient: on OpenShift the Route 502s for ~10s after the Service says Ready, and an agent whose MCP is not yet serving flaps Ready → CrashLoopBackOff → Ready.

Option	Default	Meaning
`--stable N`	4	consecutive ready polls required before proceeding
`--settle S`	15, or 45 with `--preset`/`--plugin`	pause after ready; longer when a sidecar is injected
`--poll-interval S`	10	gap between status polls
`--wait-timeout S`	1800	give up waiting for readiness

Probing MLflow. `--no-round-trip` restricts `mlflow-health` to the `auth` and `read` stages, for when you want a cheap check and not the one synthetic trace the write stages emit.

Results. `--run RUN_ID` selects the run for `poll` / `report` / `artifacts`. `--mirror DIR` downloads every artifact under `DIR` and lists them — the artifact URLs are public, so no AWS credentials are involved.

Connection and auth. Each of these takes a flag or an environment variable; the env form is usually easier because the same values drive `reference/run-12.py`.

Flag	Env	Default
`--base`	`BM_BASE`	*(required)* Service base URL
`--iss`	`BM_ISS`	*(required)* the instance's **own** issuer — the Keycloak on the cluster the Service runs on
`--user`	`BM_USER`	`benchmarker`
`--client`	`BM_CLIENT`	`rossoctl`
`--insecure`	`BM_INSECURE=1`	off — set it for OpenShift's self-signed edge routes

Two settings are **environment-only**, with no flag:

- **`BM_PASSWORD_FILE`** (a `chmod 600` file) or `BM_PASSWORD`. One is mandatory; the CLI exits rather than prompting. Prefer the file so the secret never reaches your shell history.

Exit codes. `0` success · `8` `mlflow-health` found a stage that failed · `7` the run reached a terminal state that was **not** `succeeded` · `6` a teardown that returned neither 204 nor 404 · `1` a usage or configuration error. A run that legitimately scores `pass_rate 0.0` still exits `0` if its status is `succeeded` — appworld does this by design, so do not treat exit 0 as "the agent solved it".

7.3 Other validated flows

These map to the canonical parameterized runs and were validated on `ykt3` with S3 export.

```
# Run #2 — gsm8k, 10 tasks (validated: 9/10 pass)
curl -s -X POST "$SVC/benchmarks/gsm8k/runs" -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"max_tasks": 10, "max_parallel_sessions": 1, "timeout_seconds": 600}'
# Run #3 — gsm8k, 50 tasks, 4-way concurrency (validated: 42/50, ~190s)
curl -s -X POST "$SVC/benchmarks/gsm8k/runs" -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"max_tasks": 50, "max_parallel_sessions": 4, "timeout_seconds": 900}'
# Run #9 — tau2, 10 tasks (multi-turn; deploy tau2 first)
curl -s -X POST "$SVC/benchmarks/tau2/deploy" -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" -d '{"namespace": "team1"}'
curl -s -X POST "$SVC/benchmarks/tau2/runs" -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"max_tasks": 10, "max_parallel_sessions": 1, "timeout_seconds": 1200}'
# Run #10 — tau2, 20 tasks, 4-way concurrency (validated: 17/20, ~915s — needs raised timeout)
curl -s -X POST "$SVC/benchmarks/tau2/runs" -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"max_tasks": 20, "max_parallel_sessions": 4, "timeout_seconds": 1800}'
```

7.4 The whole 12-run matrix in one command (`reference/run-12.py`)

Same idea as §7.1, one level up: instead of a single benchmark run, this drives all **12 canonical parameterized runs** — gsm8k at three sizes, a model swap, the four AuthBridge presets, tau2 at two sizes, appworld at two — deploying each leg fresh, mirroring every artifact, and writing one state file the report generators read.

Unlike `autobench-cli`, this one **is** repo-only: it is a matrix driver tied to `run12_specs.json`, not something you install. Clone first. (For the same 12 legs written out as 12 separate one-line commands, see §7.5.)

```
git clone https://github.com/rossoctl/autobench && cd autobench
# Same four variables as §7.1 — plus a label, which names the state file and the reports.
export BM_BASE=https://autobench-rossoctl-system.apps.ykt3.example.com
export BM_ISS=https://keycloak-keycloak.apps.ykt3.example.com/realms/rossoctl
export BM_PASSWORD_FILE="$HOME/.rossoctl-ykt3/benchmarker.pass"
export BM_INSECURE=1
export BM_LABEL=ykt3-v136
# Space the legs that share prompts past the LLM gateway's completion cache, and interleave the
# cache groups so the gaps overlap real work instead of being slept through. Without these two
   the
# matrix still runs -- and its latency and output-token columns are partly replays. See below.
export BM_CACHE_GAP=900
export BM_ORDER=1,12,2,10,3,11,5,9,6,4,7,8
python3 reference/run-12.py           # ~2h05m; log it: ... 2>&1 | tee /tmp/run12-ykt3.log
```

The whole matrix on another gateway. Set `BM_LLM_API_BASE`, `BM_LLM_GATEWAY`, `BM_LLM_KEY_SECRET` and `BM_MODEL_DEFAULT` to put every leg on that gateway (§6.8).

- **Models:** each leg keeps its `BM_MODEL_<ROLE>` if one is set; every other leg gets `BM_MODEL_DEFAULT`, because the spec file's model ids belong to the gateway they were written for.
- **One combination per execution.** Compare executions with `gen-gateway-compare.py`.
- **The run record keeps each deploy request without `llm_api_base`.** It feeds the committed reports, so the label stands in for the URL.

`BM_CACHE_GAP` defaults to `0`, and that default silently costs you two columns. The gateway caches completions keyed on the request body for **~10 minutes** (measured by survival curve), and task selection is deterministic, so legs #1–#3 and #5–#8 all open with the *same* five gsm8k prompts — a later leg is served the earlier leg's completions, replaying its `usage` and measuring a cache lookup instead of a call. Input tokens and pass rates are unaffected; **per-call latency and output tokens are not comparable across those legs** unless they are spaced. `900` gives a 50% margin, only the *shortfall* is slept, and an interleaving `BM_ORDER` is what keeps the cost down: the v1.36 matrices slept **28 and 14 minutes** (OpenShift, KinD) on top of about two hours of run time — less than v1.35's ~40, because slower tasks covered more of each gap with real work — where the spec-file order would have slept about twice that. **Latency is the wrong detector for a replay** — one measured replay took 3.0 s, the same as a miss — so compare response `id`s, not durations.

What the gap does not cover: tau2's opening call, task to task inside one leg. `BM_CACHE_GAP` spaces *legs*; it cannot help where consecutive tasks collide seconds apart. Inside a task nothing can collide — every call after the first re-sends the accumulated history, so each body is strictly larger than the last. In the v1.36 pair, all 279 tasks never repeat even an input-token count among their own calls (unequal counts imply unequal bodies, so it is a proof, not just an absence). The v1.35 pair had two exceptions, appworld tasks where the agent re-sent an identical request. But **every tau2 task issues a byte-identical *first* request** — 5054 input tokens on every tau2 task, both platforms — because for tau2 the prompt text is the same for every task: the scenario lives in the MCP server's user simulator, which only speaks after the agent opens, and the task identity travels in the session metadata rather than the prompt (see One task, end to end). So one or two generations serve the leg and the rest are replays, worth **~5–6% of input, ~2–3% of output and ~2–3% of chat latency** on tau2 legs and nothing elsewhere. It is not a cross-platform bias — both clusters replay identically — so the comparison the matrix exists to make still holds; just do not read a tau2 first-call

latency as a measurement. Closing it would need a `no-cache` directive on the agent's requests, which AutoBench does not control.

You can see it in the artifacts, and the v1.36 pair shows the TTL itself. Its tau2 legs ran slowly, so each one's first calls spread over 588–833 s. The one leg that stayed inside ~10 minutes (ykt3's #10, 588 s) has a single first-call output count across all 20 tasks — one generation, replayed. The three that ran past it (638–833 s) have exactly two values each: the cached completion expired mid-leg and the next task generated a fresh one. (In the faster v1.35 pair every tau2 leg finished its first calls within 460 s; there a second value came only from a parallel opening wave racing the cache.)

It ends with a per-leg summary and the state file path:

```
[18:23:48] === done in 8896s; 12 runs -> /tmp/autobench/run12-ykt3-v136-20261012.json
[18:23:48]   #1  gsm8k     succeeded      pass=1.0   tasks=1
[18:23:48]   #12 appworld  succeeded      pass=0.0   tasks=20
[18:23:48]   #2  gsm8k     succeeded      pass=1.0   tasks=10
...
```

Then turn that state file into the two documents (all numbers derived from the mirrored artifacts — nothing transcribed):

```
# 1. the 12-run report. <date> is the date the RUNS executed, from the modal run_id prefix.
python3 reference/gen-12run-report.py /tmp/autobench/run12-ykt3-v136-20261012.json v1.36 \
  "OpenShift — ykt3, single-cluster Helm install, agent runner direct (ETE-ext gateway)" \
  results/12run-report-v1.36-20261012-ocp.md
# 2. OCP vs KinD, once both matrices exist. --gateway same: both sent their model calls to one
   gateway.
python3 reference/gen-12run-comparison.py --gateway same \
  /tmp/autobench/run12-ykt3-v136-20261012.json "OpenShift (ykt3, single-cluster)" \
  /tmp/autobench/run12-kind-v136-20261012.json "KinD (single-node local)" v1.36 \
  results/12run-comparison-v1.36-20261012.md
```

Do not mine plugin overhead out of this matrix. `reference/gen-plugin-overhead.py` does exactly that — it compares the no-plugin leg #3 against the preset legs #5–#8 — and it is kept only because its source comments record why the approach fails. The canonical matrix runs each preset **once at `max_tasks=5` in a fixed order**, which cannot separate a plugin's cost from run-order drift, and its cross-benchmark projection was falsified **in both directions** (over-estimating by 2.2× on one cluster, under-estimating by 2.2× on the other). Its per-layer figures are withdrawn, not corrected. The designed replacement is a separate 13-leg experiment with its own spec file, reversed-order replicates and n=50 — see PLUGIN_OVERHEAD.md for the findings and `reference/gen-plugin-study.py` / `gen-plugin-study-xplat.py` for the generators:

```
BM_SPECS=reference/plugin_study_specs.json BM_LABEL=pstudy-ocp-v135 \
  BM_CACHE_GAP=900 BM_ORDER=111,112,101,102,103,104,105,113,106,107,108,109,110 \
  python3 reference/run-12.py        # ~3–3.5 h, most of it cache gaps; detach it
# The judge log is what tells the analyzer how many tool calls were actually authorized —
# without it, per-preset medians mix judged and unjudged calls and mislead.
oc -n rossoctl-system logs deploy/ibac-judge --since=6h --timestamps \
  | grep -vi healthz | awk '{print $1}' > /tmp/judge-ocp.ts
python3 reference/gen-plugin-study.py /tmp/autobench/run12-pstudy-ocp-v135.json \
  reference/plugin_study_specs.json v1.35 \
  "OpenShift — ykt3, single-cluster" \
  results/12run-plugin-study-v1.35-ocp.md /tmp/judge-ocp.ts
```

Switching to KinD is the same three-line change as §7.1 plus a new label:

```
export BM_BASE=http://autobench.localtest.me:8080
export BM_ISS=http://keycloak.localtest.me:8080/realms/rossoctl
export BM_PASSWORD_FILE="$HOME/.rossoctl-kind/benchmarker.pass"
unset BM_INSECURE
export BM_LABEL=kind-v136
python3 reference/run-12.py            # ~2h08m on a single node too
```

Run the two platforms sequentially, not in parallel. Each platform has its *own* litellm deployment — hence its own completion cache and its own key table, which is why a key copied between them always 401s — but the **same upstream model providers** sit behind both. Concurrent runs therefore contend exactly where it matters and the latency numbers stop meaning anything.

Other knobs, all optional:

Variable	Default	Use
`BM_CACHE_GAP`	`0`	seconds a (benchmark, model) prompt set must rest between legs so the gateway's completion cache expires. **Set it to `900`** — the default measures replays on legs #1–#3 and #5–#8
`BM_ORDER`	the spec file's	comma-separated execution order. Interleave the cache groups (`1,12,2,10,3,11,5,9,6,4,7,8`) so `BM_CACHE_GAP` overlaps real work: ~40 min slept instead of about twice that
`BM_SPECS`	`reference/run12_specs.json`	an alternate spec file, so an experiment reuses this driver without touching the canonical matrix. Accepts a bare list of legs or `{"order": [...], "legs": [...]}`
`BM_ONLY`	all 12	comma-separated leg numbers, e.g. `BM_ONLY=7` to re-run one leg, `BM_ONLY=1,2,3` for a smoke pass
`BM_STABLE_POLLS`	4	consecutive ready polls required before a run is submitted
`BM_SETTLE_PLAIN` / `BM_SETTLE_SIDECAR`	15 / 45	post-ready settle seconds; sidecar deploys need longer
`BM_USER` / `BM_CLIENT`	`benchmarker` / `rossoctl`	non-default realm identity
`BM_PASSWORD`	—	the password inline, instead of `BM_PASSWORD_FILE`. Prefer the file (`chmod 600`): an exported variable leaks into `env`, process listings and shell history

`BM_ONLY` is how you repair a matrix without redoing it: a leg that failed can be re-run on its own (`BM_ONLY=7`) under a new `BM_LABEL`, and its entry merged into the original state file. Merge only a re-run whose tasks actually passed, not one that merely reached `succeeded`, which a run that lost every task can still do.

7.5 The 12 legs as individual `autobench-cli` commands

Each leg of §7.4's matrix is also one `autobench-cli all` line, in the shape of the §7.1 example. Use these to re-run a single leg, to bisect a failure, or to read the matrix as parameters rather than as JSON. Export the same five variables as §7.1 first; every line then needs nothing else.

#	One-line command	What it varies
1	`autobench-cli all --benchmark gsm8k --tasks 1 --timeout 300 --mirror /tmp/autobench`	baseline — the smoke test
2	`autobench-cli all --benchmark gsm8k --tasks 10 --timeout 300 --mirror /tmp/autobench`	volume, still serial
3	`autobench-cli all --benchmark gsm8k --tasks 50 --parallel 4 --timeout 400 --mirror /tmp/autobench`	volume + concurrency
4	`autobench-cli all --benchmark gsm8k --model openai/Azure/gpt-4.1 --tasks 5 --parallel 4 --timeout 300 --task-timeout 120 --mirror /tmp/autobench`	model swap
5	`autobench-cli all --benchmark gsm8k --preset auth-only --tasks 5 --parallel 4 --timeout 300 --task-timeout 120 --mirror /tmp/autobench`	AuthBridge: auth-only
6	`autobench-cli all --benchmark gsm8k --preset ibac-only --tasks 5 --parallel 4 --timeout 300 --task-timeout 120 --mirror /tmp/autobench`	AuthBridge: ibac-only
7	`autobench-cli all --benchmark gsm8k --preset full --tasks 5 --parallel 4 --timeout 300 --task-timeout 120 --mirror /tmp/autobench`	AuthBridge: full (enforce)
8	`autobench-cli all --benchmark gsm8k --preset full --plugin ibac:observe --tasks 5 --parallel 4 --timeout 300 --task-timeout 120 --mirror /tmp/autobench`	full + per-plugin override
9	`autobench-cli all --benchmark tau2 --tasks 10 --timeout 2100 --task-timeout 600 --mirror /tmp/autobench`	multi-turn + user simulator
10	`autobench-cli all --benchmark tau2 --tasks 20 --parallel 4 --timeout 2400 --task-timeout 600 --mirror /tmp/autobench`	multi-turn under concurrency
11	`autobench-cli all --benchmark appworld --model openai/gemini-2.5-pro --tasks 5 --timeout 3300 --task-timeout 600 --mirror /tmp/autobench`	long-horizon
12	`autobench-cli all --benchmark appworld --model openai/gemini-2.5-pro --tasks 20 --parallel 4 --timeout 3600 --task-timeout 600 --mirror /tmp/autobench`	long-horizon under concurrency

These are generated from `reference/run12_specs.json`, not transcribed: each line's run body round-trips to the spec's `run` object field for field, and `--model` / `--preset` / `--plugin` reproduce the spec's deploy body as well (`--preset` implies `authbridge_enabled: true`; the CLI also sends `experiment: "default"`, which is the server's default anyway). Leg #1 is the §7.1 example unchanged. Two details the table encodes silently:

- **Every line is a fresh deploy.** `all` pre-cleans with `DELETE …/deploy` before deploying, which is what the matrix requires. That is also why `--no-deploy` has no place here. Warm reuse no longer drops the usage-bearing span — that agent defect is fixed upstream, measured. What a fresh deploy buys is a **fresh image pull**: these workloads are pinned to `:latest` under `imagePullPolicy: Always`, so a newly created pod picks up an upstream fix while a long-lived one serves a six-week-old digest indefinitely. It does *not* protect you from the LLM gateway's own response cache, which is outside the agent and unavoidable from here. See `docs/exgentic-agent-bug-report-20260901.md`.
- **`--settle` needs no value.** It defaults to 45 s when `--preset` or `--plugin` is present and 15 s otherwise, matching `BM_SETTLE_SIDECAR` / `BM_SETTLE_PLAIN` in §7.4.

Run them one at a time, in order. Legs sharing a benchmark share one deployment slot, so two of these in parallel will fight over it; and legs #3 and #5–#8 are only comparable because they execute byte-identical work, which a concurrent gateway load would spoil. Each exits `0` on `succeeded` and `7` otherwise (§7.2), so `&&` chaining stops at the first bad leg.

> These do **not** replace `reference/run-12.py`. The driver writes `/tmp/autobench/run12-<BM_LABEL>.json`, and that state file is the only input the three report generators accept — run the legs by hand and you get the artifacts but none of the documents. Use §7.4 for a real matrix; use this table to re-run or inspect one leg of it.

---

8. Known limits & error codes

Situation	Code	What to do
Missing/invalid bearer	`401`	fetch a fresh token (§3)
`iss` not in any instance file	`403`	add/repair the instance config (§4.1)
`/config` as non-benchmarker	`403`	authenticate as the `benchmarker` user
Setting a workload cred via `/config`	`422`	provision it as a cluster Secret instead (§4.3)
`plugin_preset`/`plugins`/`on_error` without `authbridge_enabled=true`	`422`	set `authbridge_enabled=true` so the sidecar is injected for the pipeline to take effect
`plugin_config_file` on deploy	`422`	local-path input with no HTTP analog; use `plugin_preset`/`plugins`/`on_error` instead
`llm_api_base` without `llm_gateway`, a malformed URL, Secret name or label, or an unknown deploy field	`422`	name the gateway; fix the value; check the field name (§6.1)
Run while not Ready on another gateway	`424`	the message names the deployment's `llm_key_secret`: create that Secret (§6.8)
Run before deploy	`409`	`POST …/deploy` first
Run while not Ready	`424`	provision the named Secret(s), wait for Ready (§6.2)
Report with MLflow unconfigured	`409`	set MLflow read creds via `PUT /config` (§4.5), then confirm with `GET /mlflow/health` (§5.1)
Upstream Rossoctl/MLflow failure	`502`	transient upstream issue; retry

AuthBridge plugin presets (runs 4–8 of the ibac comparison): layer-3 plugin composition and `on_error` selection require overlaying the per-agent `authbridge-config-<agent>` ConfigMap via kubectl — which the HTTP-only Service cannot do — so those are honestly rejected with `422`. The one enactable knob is `authbridge_enabled=true` (injects the sidecar with the cluster-default pipeline).

appworld: registers and deploys correctly via the Service, but its MCP image hangs on its own per-task Venv-service health timeout (120s), independent of the Service. Treat appworld as externally blocked until that image is fixed.

---

9. Extending the catalog (adding or changing a benchmark)

The benchmark catalog is **code, not runtime data** — a static `BENCHMARKS` dict in `src/autobench/benchmarks/registry.py`. There is no database, config file, or admin API behind it, and the HTTP surface over the catalog is **read-only** (`GET /benchmarks`, `GET /benchmarks/{name}`). Adding or changing a benchmark is therefore a **source change + tests + image rebuild + redeploy**, deliberately: the definition pins container images, dataset env, and resource limits that must move in lockstep with the workload images, so every change is a reviewable, git-tracked, image-versioned artifact rather than mutable state that could drift per instance.

> See the co-located contributor note `src/autobench/benchmarks/README.md` for the field-by-field reference and the per-benchmark env gotchas.

9.1 What a definition holds

Each entry is a `BenchmarkDefinition`: `name`, `mcp_image` (+ `mcp_image_tag` / `mcp_port` / `mcp_path`), `tool_env`, `tool_resources`, `default_model`, `user_simulator` (multi-turn flag), and an `agents` map of `BenchmarkAgentSpec` (per-agent `container_image` + `extra_env` + `resources`). Secret references are declared with the `_secret_env(env, secret, key)` helper; `required_secrets()` derives the actionable `424` precheck message from them automatically.

9.2 Add a new benchmark

1. **Add a `BenchmarkDefinition` entry** to `BENCHMARKS` in `registry.py`:
   - `mcp_image` (+ tag/port/path) — the MCP tool image.
   - `tool_env` — `BENCHMARK_NAME` and any secret refs via `_secret_env`. The LLM base is **not** baked in: it comes from the instance's required `workload_llm.api_base` — whichever LLM or LiteLLM service that cluster is configured to reach; a deploy with no gateway configured is rejected with 422).
   - `agents={...}` — one `BenchmarkAgentSpec` per flavor. The `tool_calling` A2A image is shared across all current benchmarks; usually you reuse it and only the name gets a `-<benchmark>` suffix.
   - `user_simulator=True` **only** for multi-turn benchmarks — this is what makes `build_tool_request` inject `EXGENTIC_SET_BENCHMARK_USER_SIMULATOR_MODEL` into the MCP pod so the server-side simulator shares the run's model.
   - Any benchmark-quirk env — mind the documented gotchas: gsm8k needs `EXGENTIC_SET_BENCHMARK_RUNNER=direct`; tau2 needs `EXGENTIC_SET_BENCHMARK_ACTION_TIMEOUT=1000`; appworld **rejects** that same action-timeout override and crashes at startup if it's present.
2. **Add tests** in `tests/test_benchmarks.py` — assert `build_tool_request` / `build_agent_request` emit the expected images, env, secrets, and resources, mirroring the existing patterns.
3. **Rebuild + bump the image tag**, redeploy the Service. The new benchmark then appears in `GET /benchmarks` and is deployable/runnable at `/benchmarks/<newname>/…` with **no client change** — the request bodies are benchmark-agnostic (see §6).

9.3 Change an existing benchmark

Same mechanism — edit the entry (bump `mcp_image_tag`, change `default_model`, add an env var, adjust `tool_resources`), update tests, rebuild, redeploy. Because `required_secrets()` is derived from `tool_env` + the chosen agent's `extra_env`, changing a secret reference automatically updates the `424` "this benchmark requires secret(s): …" precheck message — no separate wiring.

9.4 What stays runtime-mutable (for contrast)

Per-instance state is *not* in the catalog and does not require a rebuild: instance config (files under `settings.instances_dir`, keyed by `iss`) and the benchmarker-only `PUT /config` overrides (MLflow read creds + S3). The benchmark catalog is intentionally the immutable, version-pinned part.

10. Installing the Service itself

Everything above assumes a Service already answering on a hostname. Getting it there is its own document: **`ADMIN_GUIDE.md`** — prerequisites, the environment variables, the Helm chart for KinD and OpenShift, and the verification that proves the install rather than the pod.

The short version, for orientation:

- **Prerequisite on either install path: Rossoctl v0.8.0 or later.** Below that the operator accepts the deploy request the Service sends and silently drops fields it does not know, so the benchmark runs in a shape nobody configured.
- Two paths, both supported and checked against each other: the Helm chart at `deploy/helm/autobench` and the raw manifests under `deploy/`. `python3 reference/helm-parity-check.py` asserts they render the same objects.
- Before installing (and again afterwards, as an audit), `python3 reference/preflight.py` checks the cluster prerequisites read-only and reports every credential as a truncated hash.
- The `autobench-instances` Secret — one JSON file per issuer, mounted at `/etc/service/instances` — is **never created by an install path**. It holds ROPC credentials and is generated out-of-band by `reference/{kind,ocp}-service-bootstrap.sh`. The Service reads instance files at startup, so `rollout restart` after changing it; `PUT /config` (§5) is in-memory only.

11. Regenerating the documents

Everything under `docs/` that is not hand-written has a generator, so no step in producing the published artifacts is a manual browser print or a hand-edited table.

Artifact	Generated by	Source of truth
`docs/DEVELOPER_GUIDE.pdf`, `docs/ADMIN_GUIDE.pdf`	`python3 reference/gen_pdf.py`	the `.md` beside it
the table of contents in this file (and in the Admin Guide)	`python3 reference/gen_toc.py docs/DEVELOPER_GUIDE.md`	that file's own headings
§7.5's 12 one-line commands	`reference/run12_specs.json`	the same specs `reference/run-12.py` executes

Artifact	Generated by	Source of truth
§6.8's cb vs ETE-Ext table and its two figures (`docs/img/gateway-*.png`)	`uv run --with matplotlib python reference/gen-gateway-legs.py <run.json> "ETE-Ext" <run.json> "cb"`	each leg's mirrored `run.json` + `report.ndjson`, and `model_prices.json` per gateway
the tool-selection table in §2 and in the primer	`python3 reference/gen-shortlist-audit.py <run12.json> "<label>" …`	`span_report.ndjson` in each leg's mirror
`docs/AutoBench.pptx`, `docs/AutoBench.pdf`	`docs/deck/.venv/bin/python docs/deck/make_autobench_pptx.py` (PDF via PowerPoint; `DECK_PDF_VIA=soffice` for LibreOffice)	`docs/results/v1.36-2026-10-12/` (12-run), `docs/results/v1.35-2026-10-04/` (plugin study), `docs/img/takeaways.json` + this guide

The deck has its own venv, never the project's: `python3 -m venv docs/deck/.venv && docs/deck/.venv/bin/pip install -r docs/deck/requirements.txt`. Edit the generator, never the `.pptx`; a hand edit is lost at the next build.

`gen_pdf.py` with no arguments rebuilds every PDF in its `PAIRS` list (~10 s). It dispatches on extension: `.md` goes through pandoc → CSS → headless Chrome, an ad-hoc `.pptx` through LibreOffice. Pass paths to do a subset, `-o` to write elsewhere:

```
python3 reference/gen_pdf.py                                  # every pair in PAIRS
python3 reference/gen_pdf.py docs/DEVELOPER_GUIDE.md          # just this guide
python3 reference/gen_pdf.py docs/ADMIN_GUIDE.md -o /tmp/draft.pdf
```

Every generated PDF carries a bookmarks outline (the navigation sidebar), and each run prints what it produced — `outline: 36 bookmarks, depth 0–3, first='…'` — so a silent regression to a PDF you cannot navigate shows up in the build output instead of in a reader's lap. Two different mechanisms produce it, which is why both needed fixing:

- `.md` → **Chrome** builds the outline from the `h1`–`h6` tree, but only with `--generate-pdf-document-outline` (Chrome ≥ 122). Without that switch Chrome emits no outline at all, and it ignores the unknown flag silently on older builds.
- **The deck** gets its bookmarks from `make_autobench_pptx.py` itself: neither PowerPoint nor LibreOffice writes useful ones, so the build rewrites them with pypdf from its `SECTIONS` list ("1. Section", its slides beneath), and stops if the page count differs from the slide count.

The CSS lives in the script rather than in a theme file for one reason worth knowing before you change it: **long code lines must wrap, not clip.** A 190-char `autobench-cli` line or a 185-char S3 URL runs off the page under default print styling, and the flags that matter are at the end. Three rules prevent that — `white-space: pre-wrap`, a hanging indent per *logical* source line (so a wrapped continuation is visibly a continuation and not a second command), and `overflow-wrap` where there are no spaces to break at. Long table cells opt into breaking mid-word by measured length; applying it to every cell shrinks narrow columns until two-digit numbers stack vertically.

Every artifact in `docs/` is covered by the table above, and it is worth keeping that way: if no generator rebuilds it, don't commit it. A hand-made export does not merely go out of date — it keeps publishing what the current text has retracted, and nothing in the build output will tell you. That is why there is no Word copy of this guide and no rendered PDF of the OpenAPI documents. If you need another format, convert on

demand (`pandoc docs/DEVELOPER_GUIDE.md -o /tmp/guide.docx`) and leave the result out of the repo. The OpenAPI documents themselves — `docs/openapi.json` / `docs/openapi.yaml` — are dumps of the schema `_install_openapi()` builds in `src/autobench/app.py` rather than hand-written text, but no dump script is committed either: refresh them from a running Service's `/openapi.json`.